

Deep Reinforcement Learning for On-Orbit Servicing Scheduling for Low and High Thrust Propulsion

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Abstract

Deep Reinforcement Learning for On-Orbit Servicing Scheduling for Low and High Thrust Propulsion

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The increasing density and economic value of satellite infrastructures make on-orbit servicing (OOS) an important capability for future space operations. This thesis presents the Integrated On-Orbit Services Scheduling Optimization (IOOSSO) framework, a deep reinforcement learning approach for dynamically scheduling OOS missions in multi-orbit environments. The problem is formulated as an event-driven semi-Markov decision process with stochastic service needs, orbital refuelling depots, and fuel, time, and orbital-transfer constraints. High-thrust (HT) and low-thrust (LT) architectures are modeled within a continuous orbital representation, with feasible rendezvous transfers constructed on the fly. A Maskable Proximal Policy Optimization agent learns servicing policies that maximise an integrated-system net present value (NPV) by balancing preserved fleet revenue against infrastructure, propellant, and asset-loss costs.

Medium Earth Orbit case studies show that servicing profitability is strongly horizon- and demand-dependent. Neither architecture recovers its infrastructure cost over one year, whereas both break even within the first year of a five-year campaign and create approximately \$500M of value relative to a replace-on-failure baseline. LT is favored at low demand because of its propellant efficiency, but HT overtakes it at about 34 satellites as transfer speed becomes decisive. Additional depots create value only when constellation geometry makes inter-orbit manoeuvres expensive, and optimized placement can nearly substitute for depot count. Finally, reinforcement learning provides decisive gains for LT and under constrained refuelling logistics, while a greedy heuristic remains competitive for HT when transfers are short and depot access is benign.

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List of Acronyms

- OOS : On-Orbit Servicing
- NASA : National Aeronautics and Space Administration
- ESA : European Space Agency
- LEO : Low Earth Orbit
- MEO : Medium Earth Orbit
- GEO : Geosynchronous Earth Orbit
- LT : Low Thrust
- HT : High Thrust
- OTV : Orbital Transfer Vehicle
- RL : Reinforcement Learning
- DRL : Deep Reinforcement Learning
- MDP : Markov Decision Process
- PPO : Proximal Policy Optimization
- TSP : Travelling Salesman Problem
- NPV : Net Present Value
- IOOSSO : Integrated On-Orbit Services Scheduling Optimization

Nomenclature

Sets, Entities, and Indices

- $\mathcal{O}$: set of Earth orbits
- $\mathcal{S}$: set of client satellites
- $\mathcal{D}$: set of orbital refuelling depots
- $\mathcal{S}_t^{\text{alive}}$: set of unserviced, operational, and still-aging satellites at time step t
- $\mathbb{S}$: state space of the decision process
- $\mathbb{A}$: complete action space
- $\mathcal{M}_t$: set of admissible actions at decision epoch t
- o : orbit belonging to $\mathcal{O}$
- s : client-satellite index
- d : refuelling-depot index
- v : servicing vehicle
- j : candidate target entity
- a : transfer-arc index

Time and Decision Process

- t : discrete simulation time-step index
- T : mission horizon, expressed in simulation time steps
- Δt : duration of one simulation time step
- k : decision-epoch index
- K : total number of decision epochs in an episode
- t_k : simulation time corresponding to decision epoch k
- R_k : reward accumulated between decision epochs k and $k + 1$
- H : preventive refuelling anticipation window
- T_{yr} : duration of one year, taken as 365.25 days

Orbital Mechanics and Transfer Model

- θ : true anomaly
- $\Delta\theta$: true-anomaly difference between the servicer and a target
- μ : standard gravitational parameter of the central body
- n : orbital mean motion
- n_1, n_2 : mean motions of the departure and arrival orbits
- r : circular-orbit radius
- r_0 : reference circular-orbit radius
- a_{ph} : semi-major axis of the phasing orbit
- T_{ph} : orbital period of the phasing ellipse

- t_w : pre-transfer coasting time required for rendezvous
- t_H : Hohmann transfer time
- T_{syn} : synodic period between two orbits
- Δi : orbital inclination change
- v : circular orbital velocity
- v_1, v_2 : circular velocities of the initial and final orbits
- $\Delta v, \Delta V$: velocity increment required by an orbital manoeuvre
- Δv_{ph} : velocity increment of a high-thrust phasing manoeuvre
- Δv_{pc} : velocity increment of a plane-change manoeuvre
- τ : transfer duration
- τ_a : duration of transfer arc a
- ψ_a : velocity increment associated with transfer arc a
- p_v : current orbital position of the servicer
- p_j : current orbital position of target j

Servicer and Propulsion Model

- f_t : servicer remaining onboard propellant mass at time step t
- F_{HT} : maximum onboard propellant capacity of the high-thrust servicer
- F_{LT} : maximum onboard propellant capacity of the low-thrust servicer
- ρ_a : propellant mass consumed along transfer arc a
- m : current servicer mass

- I_{sp} : propulsion-system specific impulse
- n_{isp} : number of selectable specific-impulse operating points
- g_0 : standard gravitational acceleration
- c : effective exhaust velocity, $c = I_{sp}g_0$
- P : electrical power available to the low-thrust propulsion system
- η : low-thrust propulsion efficiency
- a_{th} : low-thrust acceleration
- $\dot{m}$: propulsion-system mass flow rate
- α : coasting fraction of the low-thrust bang–coast–bang phasing profile

Satellite Reliability and Service Demand

- $o_{s,t}$: cumulative operating age of satellite s at time step t
- $\bar{o}^r$: operating-age threshold triggering a refuelling need for satellite s
- F_s^m : Weibull cumulative distribution function for maintenance need occurrence
- F_s^d : Weibull cumulative distribution function for deorbiting failure occurrence
- β_s^m : Weibull shape parameter for maintenance degradation of satellite s
- β_s^d : Weibull shape parameter for deorbiting failure of satellite s
- η_s^m : Weibull scale parameter for maintenance degradation of satellite s
- η_s^d : Weibull scale parameter for deorbiting failure of satellite s
- p_s^m : one-step probability of a maintenance need for satellite s
- p_s^d : one-step probability of a deorbiting failure for satellite s

- $y_{s,t}$: operational indicator of satellite s at time step t
- $P_s^d(t)$: probability of a deorbiting failure over the remaining evaluation window

Reward and Economic Model

- r_s : annual revenue generated by satellite s
- $\bar{r}_s$: revenue generated by satellite s during one simulation time step
- Π_s : asset or replacement value of satellite s
- r_t : total immediate reward at time step t
- r_t^{rev} : forgone-revenue component of the reward
- r_t^{risk} : expected asset-loss component of the reward
- r_t^{fuel} : propellant-cost component of the reward
- r_t^{of} : out-of-fuel failure penalty
- R : cumulative episode return
- c^{fuel} : propellant unit price
- I_0 : initial servicing-system investment
- I_0^{depot} : depot capital expenditure allocated to the mission
- c_{sk} : daily station-keeping cost per refuelling depot
- δ : annual financial discount rate
- $D(t)$: financial discount factor at time step t
- CF_t : net operating cash flow at time step t
- NPV: net present value of the servicing system

- Π_{depot} : acquisition price of one orbital refuelling depot
- L_{depot} : design lifetime of a refuelling depot
- n_{depots} : number of orbital refuelling depots
- $C_{\text{repl}}(t)$: prorated satellite replacement cost at time step t
- L_{nom} : nominal design lifetime of a client satellite
- $V_s(t)$: value-at-stake score of satellite s used by the greedy heuristic

Reinforcement Learning

- π : servicing policy
- π^* : policy maximizing the expected cumulative return
- π_θ : policy parameterized by neural-network parameters θ
- θ_{old} : policy parameters before the current PPO update
- L^{CLIP} : PPO clipped surrogate objective
- $\varrho_t(\theta)$: PPO importance-sampling probability ratio
- $\hat{A}_t$: estimated advantage at time step t
- ϵ : PPO clipping parameter
- $\mathbf{z}_t$: vector of raw policy-network logits
- $z_{t,a}$: raw policy logit associated with action a
- γ : reinforcement-learning discount factor

Chapter 1

Introduction

Over the past decade, Earth orbit has experienced sustained and rapid growth in the number of operational satellites, driven by major advances in spacecraft miniaturization, mass production, and launch capabilities. Reusable launch systems and increasingly accessible commercial launch services have substantially reduced the cost of access to space, enabling the deployment of large constellations across all major orbital regimes, including Low Earth Orbit (LEO), Medium Earth Orbit (MEO), and Geosynchronous Earth Orbit (GEO) (Jones, 2018). This trend is reshaping the orbital environment: as of March 2026, nearly 15,000 active satellites were in orbit, while the number and scale of commercial constellations continue to increase (ESA Space Debris Office, 2025). Orbital infrastructures are therefore becoming simultaneously more valuable and more congested, making satellite availability, resilience, and long-term sustainability increasingly important.

In parallel with this expansion, on-orbit servicing (OOS) has emerged as a strategic capability for future space operations (Flores-Abad, Ma, Pham, & Ulrich, 2014). Rather than treating satellites as disposable assets, OOS seeks to extend mission lifetimes and preserve service continuity through in-space intervention. Depending on the mission context, such interventions may include refuelling, maintenance, repair, inspection, upgrading, relocation, or end-of-life disposal. The ability to restore or preserve spacecraft functionality can reduce replacement costs and service interruptions while limiting the accumulation of defunct objects in orbit. These benefits are particularly relevant in heavily utilized orbital regions, where failed spacecraft may remain in orbit for extended periods and become hazards to other operators (ESA Space Debris Office, 2025).

However, planning OOS missions is intrinsically difficult. Satellite service needs may emerge unpredictably as spacecraft degrade and fail, while all orbital assets continuously evolve along their respective trajectories. The feasibility, duration, and propellant cost of a rendezvous depend on the current positions of both the servicer and its target. Each servicing decision must therefore account not only for the immediate value of an intervention, but also for the opportunity cost of the transfer, the needs that may emerge during transit, and the propellant that must remain available for future operations. These trade-offs become particularly important when a single servicer operates across multiple orbital planes and altitudes, and when access to orbital refuelling infrastructure is limited. OOS scheduling is therefore a sequential decision-making problem under uncertainty in which locally attractive decisions may have substantial long-term consequences.

Existing OOS planning studies have largely relied on static mathematical programming, meta-heuristics, or rolling-horizon optimisation. Although these methods can provide effective servicing plans for known mission instances, stochastic degradation and evolving service demand require the plan to be repeatedly reconsidered as new information becomes available. Reinforcement learning (RL), by contrast, offers a natural framework for learning adaptive decision policies that map the current operational state directly to servicing decisions. Recent RL studies have demonstrated this potential for related orbital sequencing problems, particularly active debris removal. However, as discussed in Chapter 2, comparatively little attention has been devoted to RL-based OOS scheduling with explicit economic objectives, stochastic satellite degradation, orbital refuelling infrastructure, and a direct comparison of high-thrust and low-thrust propulsion architectures.

To address this gap, this thesis introduces the Integrated On-Orbit Services Scheduling optimisation (IOOSSO) framework, a deep reinforcement learning approach for dynamically scheduling OOS missions in a multi-orbit environment. The servicing problem is formulated as an event-driven semi-Markov decision process in which a servicing spacecraft selects, at each decision epoch, the next client satellite or refuelling depot to visit. In the low-thrust configuration, the servicer additionally selects its propulsion operating point. Decisions are made while accounting for the servicer’s remaining propellant, the current positions and operational conditions of the client satellites, emerging service needs, transfer durations, and the accessibility of refuelling infrastructure.

The IOOSSO framework uses a continuous orbital representation. Client satellites, refuelling

depots, and the servicer are described by their current orbital positions, and feasible rendezvous transfers are constructed on the fly from the state of the environment at each decision epoch. Both high-thrust (HT) and low-thrust (LT) propulsion architectures are explicitly modeled. The HT architecture provides rapid transfers at the expense of comparatively high propellant consumption, whereas the LT architecture offers substantially greater propellant efficiency but may require weeks or months to reach a target. The framework therefore makes it possible to investigate how the fundamental time–fuel trade-off between the two propulsion modes affects servicing performance as demand and mission conditions evolve.

Satellite service needs are modeled through deterministic refuelling deadlines and stochastic degradation and failure processes. The servicing agent must consequently adapt its trajectory to a fleet whose condition changes throughout the mission. A Maskable Proximal Policy optimisation agent is employed to solve the resulting constrained decision problem. Invalid-action masking excludes targets and propulsion settings that are infeasible in the current state, particularly when a transfer exceeds the servicer’s available propellant. The learning objective is economically grounded: the decision-level reward is aligned with the net present value of the integrated servicing system by accounting for fleet revenue forgone during satellite outages, expected irreversible asset losses, and propellant expenditure. The resulting policy is therefore trained to protect the long-term economic value of the serviced fleet rather than simply minimize transfer cost or maximise the number of completed services.

In addition to assessing the performance of the learned policy, this thesis investigates when deep reinforcement learning is actually necessary for OOS scheduling. The Maskable PPO policy is therefore benchmarked against a transparent, training-free greedy value-at-stake heuristic operating under the same orbital dynamics and feasibility constraints. This comparison distinguishes scenarios in which anticipatory, horizon-level optimisation creates meaningful value from those in which informed myopic decision-making is sufficient. More broadly, the proposed framework is used to study three interconnected design questions: how the relative economic value of HT and LT propulsion evolves with service demand, when orbital refuelling depots justify their infrastructure cost, and under which operational conditions reinforcement learning provides a decisive scheduling advantage.

The framework is evaluated through a set of MEO servicing case studies inspired by modern multi-orbit communication constellations. The numerical experiments vary the mission horizon, the size of the client fleet, the number and placement of refuelling depots, and the orbital geometry of the constellation. The results reveal that OOS profitability is strongly horizon-dependent: neither propulsion architecture recovers its infrastructure cost over a one-year campaign, whereas both break even within the first year of a five-year campaign and create close to \$500M of additional value relative to a replace-on-failure baseline. The relative advantage of the two propulsion architectures is similarly demand-dependent. LT is slightly favored for small fleets because of its high propellant efficiency, but HT overtakes it at approximately 34 satellites as transfer speed becomes increasingly important. At high demand, the LT servicer eventually saturates because service needs accumulate faster than its long transfers allow them to be addressed.

The results further demonstrate that refuelling infrastructure cannot be assessed independently of constellation geometry. Additional depots are economically counter-productive when the serviced orbital planes are nearly coplanar, since the reduction in transfer cost is insufficient to recover the additional infrastructure expenditure. When inclination changes become more expensive, however, depot access becomes a major operational constraint and additional infrastructure can create substantial value. Depot placement proves at least as important as depot count: in the most constrained case study, a single strategically positioned depot captures nearly the full benefit of the best two-depot configuration. Finally, the comparison of scheduling policies shows that reinforcement learning is most valuable when current decisions create long-lived commitments. It provides decisive gains for the slow LT architecture as service demand increases and for HT missions with constrained propellant logistics. Conversely, when HT transfers are short and depot access is inexpensive, the greedy value-at-stake heuristic remains competitive with the learned policy.

Taken together, these findings highlight the strong interdependence between propulsion architecture, expected service demand, orbital geometry, refuelling infrastructure, and scheduling policy. The contribution of this thesis is therefore not limited to the development of an RL scheduler. The IOOSSO framework provides a unified methodology for dynamically planning OOS missions and for economically comparing alternative system architectures under a common stochastic operating environment.

The remainder of this thesis is organized as follows. Chapter 2 reviews existing approaches to on-orbit servicing mission planning, with particular attention to static optimisation and reinforcement-learning-based orbital sequencing methods. Chapter 3 presents the IOOSSO framework, including the orbital environment, service-need model, event-driven decision formulation, economic model, Maskable PPO agent, and greedy benchmark policy. Chapter 4 evaluates the framework through a series of MEO case studies and analyzes the effects of mission horizon, service demand, propulsion architecture, refuelling infrastructure, and scheduling policy. Finally, Chapter 5 summarizes the principal findings, discusses the limitations of the present work, and identifies directions for future research.

Chapter 2

Literature Review

In this section, we will review a large part of the literature in the field of On-Orbit Servicing (OOS) and maintenance planning.

First of all, what does OOS mean and what does this term include? On-Orbit Servicing refers to all operations carried out on satellites or spacecraft already in orbit, such as repair, refuelling, inspection, or modification, in order to extend their lifetime or enhance their capabilities. The first robotic vehicles for on-orbit servicing (OOS) were designed as highly integrated spacecraft, in which all systems (propulsion, robotic arm, navigation, etc.) were contained within a single compact structure that combined all subsystem technologies into one unit, for example, the Orbital manoeuvring Vehicle (OMV) Tatsch, Fitz-Coy, and Gladun (2006) Alizadeh and Zhu (2024). The OMV was a project developed by NASA in the 1980s to move satellites, perform repairs, and more, but it was never launched. This historical absence of maintenance infrastructure has led either to tolerating degradations (partial failures, performance decline) or abandoning satellites (increasing the amount of orbital debris). Only software updates, which can sometimes extend operational use (e.g., BIRD), have been carried out in the past. All of this illustrates a general lack of on-orbit servicing capability Stoll et al. (2009).

The rise of mega-constellations in recent years has reinforced the need for on-orbit maintenance. Indeed, more than 7,000 satellites have been launched by SpaceX since 2023 (as part of the Starlink constellation), in addition to those from OneWeb and others Dremann et al. (2025). Without maintenance or controlled end-of-life procedures, the risk of debris accumulation (the Kessler syndrome)

increases. On-orbit maintenance therefore appears essential to ensure the safety and sustainability of space operations Dremann et al. (2025). Recent demonstrators, such as Northrop Grumman’s Mission Extension Vehicle (MEV), have proven the feasibility of on-orbit refuelling and robotic repair, paving the way for routine maintenance services to extend satellite lifespans and manage end-of-life satellites.

In this context, on-orbit services planning becomes a complex problem that therefore requires appropriate tools for its resolution. Two main types of solution approaches can be distinguished. On one hand, there are static methods, which develop a visiting plan in advance based on input data that remain unchanged throughout the resolution process. On the other hand, there are dynamic or adaptive methods, in which decision-making occurs during the mission according to the system’s state—often using reinforcement learning (RL) techniques to handle uncertainty, and modeling the problem as a Markov Decision Process (MDP). There also exist hybrid approaches that combine both paradigms. Therefore, in the following section, we will present the state of the art of these different methods as applied to satellite on-orbit services.

To begin with, as mentioned earlier, the majority of static on-orbit service planning methods do not take into account the uncertainties of the environment. Indeed, decisions are made solely based on the data available at a given time t , without continuous adjustment according to the actual state of the system. A first static resolution approach consists in modeling the problem as a Vehicle Routing Problem (VRP), which aims to find an optimal visiting sequence in order to minimize one or more parameters, generally the fuel consumption expressed in terms of ΔV . This approach was proposed by Shen and Tsotras (2002), where the objective is to determine the optimal sequence of rendezvous for a servicing satellite (SSc) responsible for visiting and refuelling or repairing multiple satellites located on the same circular orbit. The problem is solved using integer programming combined with a heuristic study, while minimizing the total ΔV required to complete all rendezvous within a predefined mission time.

Sorenson and Pinkley (2023) also made an important contribution by formulating the on-orbit servicing scheduling problem as a Mixed-Integer Linear Programming (MILP) model that maximises the weighted completion of servicing tasks subject to orbital transfer, refuelling, and timing

constraints. Their formulation is noteworthy in that it supports cross-regime servicing missions across LEO, MEO, and GEO through the incorporation of refuelling depots. Their incorporation of refuelling depots into cross-regime servicing missions is a feature the present work retains; our environment, however, abandons the discretized network representation in favor of a continuous orbital state in which transfers are constructed on the fly.

Other studies have investigated more complex multi-satellite scenarios using optimisation methods. For example, Han, Guo, Li, Zhi, and Lv (2022) used an Adaptive Genetic Algorithm to optimize the mission planning of multiple satellite repair operations in GEO, considering multiple servicing vehicles. In this case, the problem is a many-to-many scenario, where several servicing vehicles must visit a subset of satellites to perform repairs while minimizing the total ΔV cost and adhering to a predefined time horizon.

Du, Zhao, Dutta, Yu, and Chen (2015) also used a genetic algorithm to minimize the nonlinear fuel consumption in ΔV (based on the Tsiolkovsky equation), all modeled using a Mixed Integer Non-Linear Programming (MINLP) formulation. Dutta and Tsotras (2007) address the problem of peer-to-peer (P2P) satellite refuelling and solve it using a GRASP (Greedy Randomized Adaptive Search Procedure) algorithm with local search and path relinking to minimize the total ΔV consumption.

Ho and co-workers have also extensively studied the on-orbit servicing problem. In Choi and Ho (2025), they introduce the Continuous Location Routing Problem for Orbital Depots (CLRPOD), which jointly determines the optimal location of orbital propellant depots and the servicing routes of the spacecraft departing from them, including visits to a fixed constellation of satellites to perform refuelling before returning, while minimizing the total propellant mass, expressed in an “effective mass to LEO” (EMLEO) metric. This joint treatment of depot placement and servicing routes is the most direct precedent for the infrastructure-sizing questions examined in the present paper, albeit within a deterministic, cost-minimization setting.

Planning problems for on-orbit servicing under uncertainty may also be tackled using static optimisation methods. Sarton Du Jonchay, Chen, Gunasekara, and Ho (2021) proposed such an approach by developing a framework for the modeling and optimisation of complex OOS missions involving one or several servicers and orbital refuelling depots under uncertain service demands.

Their formulation considers low-thrust, high-thrust, and hybrid propulsion architectures within a unified framework. The problem is formulated as a Mixed-Integer Linear Program (MILP), and uncertainty is addressed through the use of a rolling-horizon solution strategy. Together with the CLRPOD, this framework, which provides a unified treatment of high-thrust, low-thrust, and hybrid architectures interacting with orbital refuelling depots under uncertain demand, constitutes the direct foundation on which the present study builds.

To conclude, static optimisation approaches are attractive because they can provide an optimal maintenance plan for a fully specified deterministic instance. Nevertheless, their inherent limitation lies in their inability to naturally accommodate the stochastic occurrence of failures and unexpected events in real time. In practice, addressing uncertainty within such frameworks typically requires the use of replanning strategies, such as rolling-horizon methods, which introduce a significant computational trade-off. This is a major limitation for on-orbit servicing applications, since the orbital environment is highly dynamic and affected by numerous uncertainties that must be incorporated into the decision process to achieve resilient and operationally effective servicing policies.

In contrast to static methods, dynamic resolution methods make maintenance decisions in an adaptive manner. Indeed, a dynamic model relies on the current state of the environment (such as the condition of the satellites) to make the best possible decision. In this context, reinforcement learning has emerged as an essential approach. In fact, an increasing number of studies have sought to demonstrate the effectiveness of RL in sequential decision-making optimisation under uncertainty. In general, dynamic decision-making problems are modeled as Markov Decision Processes (MDPs), in which an agent interacts with an environment. At each step, the agent selects an action, and the environment responds by providing a new observation of the system's state, along with a reward associated with the previous action, indicating whether it had a positive or negative effect. The goal of the agent is to make the best possible decisions in order to maximise the cumulative reward, for example, maximizing satellite availability or minimizing maintenance costs over its lifetime.

Beyond general maintenance planning, reinforcement learning has also been applied directly to orbital mission sequencing problems, particularly in the context of active debris removal (ADR). ADR missions share strong structural similarities with the on-orbit servicing scheduling problem

addressed in this thesis: both involve a spacecraft that must visit a set of orbital targets under fuel, time, and transfer constraints, and both can be formulated as variants of the time-dependent Travelling Salesman Problem (TSP). The literature on RL-based ADR mission planning is therefore highly relevant to the present work and is reviewed below.

Yang, Hou, Hu, Liu, and Pan (2020) proposed one of the earliest reinforcement learning formulations for active debris removal mission planning. In their work, the ADR problem is modeled as a maximal-reward optimisation problem in which an Orbital Transfer Vehicle (OTV) must select a removal sequence among a set of LEO debris objects so as to maximise the total number of removed targets, subject to ΔV and mission duration constraints. The authors recast this optimisation model into an RL scheme, defining state, action, and reward components consistent with the MDP framework, and solved it using a modified Upper Confidence bound Tree (UCT) search algorithm. Their modified UCT retains the roll-out simulation from classical Monte Carlo Tree Search to provide robust state-value estimates during the early stages of training, while incorporating neural-network-assisted selection and expansion procedures to improve exploration efficiency.

Bandyopadhyay and Waxenegger-Wilfing (2024) introduced a masked Proximal Policy optimisation (PPO) algorithm for determining the most time-efficient sequence of debris visitation in a multi-debris rendezvous scenario. The problem is formulated as a modified TSP in which a spacecraft must rendezvous with a set of debris objects, using Izzo’s adaptation of Lambert’s problem to compute individual transfer trajectories. The RL agent operates in a discrete action space corresponding to the selection of the next debris target, while invalid actions, namely attempts to revisit already-serviced debris objects, are excluded through action masking.

In a subsequent study, Bandyopadhyay and Waxenegger-Wilfing (2026) extended their framework to incorporate refuelling logistics and adaptive collision avoidance into the RL-based mission planning pipeline. Collision risks are modeled as cuboidal danger zones that appear stochastically during target selection, requiring the agent to replan its trajectory using elliptical detour manoeuvres with a minimum clearance of 5 km. refuelling is introduced as a learnable decision: the agent may choose to refuel at designated checkpoints, but only after completing at least one successful rendezvous.

Finally, Tomanek-Volynets, Ceriotti, and McInnes (2024) addressed the multi-target sequence

optimisation problem using a Graph Pointer Network (GPN) trained with the REINFORCE algorithm. Their approach targets scenarios with a large number of orbital objects (up to 200 targets) and focuses on minimizing the total ΔV required for a round-trip mission, with individual transfers computed via optimized Lambert arcs. The GPN, a sequence-to-sequence neural network augmented with graph embedding layers, outputs a probability distribution over candidate targets at each decision step, with already-visited targets masked to enforce feasibility.

Collectively, these studies confirm the growing potential of deep reinforcement learning for sequential orbital mission planning under combinatorial and dynamic constraints. Two strands of this literature come closest to the present work, and together they delimit its contribution. On the methodological side, Bandyopadhyay and Waxenegger-Wilfing (2024, 2026) employ the same algorithmic machinery as this paper, i.e., proximal policy optimisation with invalid-action masking over a discrete target-selection space, later extended with refuelling stops, but direct it at time- and fuel-efficient debris rendezvous: their targets are inert objects, so their formulations involve no client fleet generating revenue, no stochastic degradation and failure economics, and no economic assessment of the servicing infrastructure itself. On the problem side, the works of Choi and Ho (2025); Sarton Du Jonchay et al. (2021) form the foundation this paper builds upon: they established the joint consideration of orbital depots and servicing routes, and the comparative evaluation of high-thrust and low-thrust propulsion for on-orbit servicing under demand uncertainty. As static optimisation frameworks, however, they plan against a fixed (or rolling-horizon) problem instance and minimize mission cost, without an integrated fleet-level economic model. The present work extends this line in two directions. First, it introduces a whole-fleet discounted net-present-value model in which satellites fail randomly as they age, and in which every scheduling decision is weighed against forgone revenue and the expected cost of failures, thereby making the value created by servicing, rather than its cost alone, both the optimisation objective and the evaluation metric. Second, it addresses the joint sizing of the propulsion architecture and the refuelling-depot network, both in number and in placement, within that economic model, evaluated through policies that adapt online to the realized failure history.

Chapter 3

Methodology

3.1 Orbital Environment Simulation

We begin by describing the orbital environment in which the reinforcement learning agent, namely the servicer, evolves and takes its decisions. The environment is composed of the following entities, each of which is detailed in the subsections below: a set of Earth orbits $\mathcal{O}$, a fleet of client satellites $\mathcal{S}$, a set of orbital refuelling depots $\mathcal{D}$, and a single servicing vehicle v .

3.1.1 Orbits

We consider a set of Earth orbits, $\mathcal{O}$, on which the fleet of client satellites that may require servicing during the mission horizon is distributed. Each orbit $o \in \mathcal{O}$ is defined by its classical Keplerian elements, namely its semi-major axis, inclination, and right ascension of the ascending node. For simplicity, all orbits are assumed to be perfectly circular. Under this assumption, any position along an orbit is fully described by a single angular coordinate, the true anomaly $\theta \in [0^\circ, 360^\circ]$, and any object placed on the orbit drifts at the constant rate given by the orbit’s mean motion. This continuous angular parameterization underpins the position representation used throughout the remainder of the framework: every entity in the environment is located by the pair (o, θ) , without resorting to any spatial discretization of the orbits.

3.1.2 Temporal Evolution

To capture the temporal evolution of the system, we introduce a discrete time horizon divided into multiple intervals: $t = 0, \dots, |T|$, representing the maximum duration available to perform service operations (e.i. the maximum time of the simulation).

3.1.3 Satellites

We define a set of client satellites, $\mathcal{S}$, where each satellite $s \in \mathcal{S}$ may require one on-orbit servicing action during the mission horizon. Each satellite is associated with a predefined orbit from the set $\mathcal{O}$ and an initial true anomaly on this orbit. Since all orbits are assumed to be circular, satellite motion is modeled deterministically within the continuous orbital representation. More specifically, the position of each satellite is updated at every simulation step according to the angular velocity of the satellite and the selected simulation time step. As a result, the true anomaly occupied by a satellite at time t is entirely determined by its initial position and its orbital progression.

In addition to its orbital position, each satellite is characterized by an operating-time state, denoted by $o_{s,t}$, which represents the cumulative operational age of satellite s at time step t . This quantity is incremented by one at each simulation step until the satellite is serviced. In the proposed framework, satellite service requirements arise from two complementary mechanisms: (i) a deterministic fuel-depletion threshold that triggers a refuelling need, and (ii) stochastic degradation processes governing the emergence of maintenance and deorbiting needs.

Let $\bar{o}_s^r$ denote the operating-time threshold beyond which satellite s is assumed to require refuelling. In the model, once

$$o_{s,t} \geq \bar{o}_s^r, \quad (1)$$

a refuelling request is activated for satellite s , provided that the satellite has not already entered a deorbiting state. This mechanism reflects the fact that propellant depletion can be approximated as a gradual and predictable aging process. Indeed, unlike the stochastic degradation processes described below, propellant exhaustion is a predictable failure mode, since the remaining fuel level of a client satellite can reasonably be assumed to be observable by its operator. To exploit this predictability, a preventive refuelling window H is introduced: the servicer is allowed to refuel satellite s up to H

days before its fuel-failure threshold is reached, that is, as soon as

$$o_{s,t} \geq \bar{o}_s^r - H, \quad (2)$$

where H is expressed in time steps consistently with the adopted time discretization. refuelling within this anticipation window prevents the upcoming service outage altogether, thereby increasing the satellite's operational availability and the revenue it generates over the mission horizon. Note that the window only defines the earliest admissible refuelling epoch; it does not act as an upper bound, and a satellite may still be refueled after its fuel-failure threshold has been exceeded.

Mechanical degradation and end-of-life failure are modeled through Weibull laws, which are widely used in reliability engineering to represent age-dependent failure processes. For each satellite s , two Weibull processes are defined: one associated with the occurrence of a maintenance need, and one associated with the occurrence of a deorbiting need. Their cumulative distribution functions are written as

$$F_s^m(o) = 1 - \exp \left[- \left(\frac{o}{\eta_s^m} \right)^{\beta_s^m} \right], \quad (3)$$

and

$$F_s^d(o) = 1 - \exp \left[- \left(\frac{o}{\eta_s^d} \right)^{\beta_s^d} \right], \quad (4)$$

where β_s^m and β_s^d are the Weibull shape parameters, and η_s^m and η_s^d are the corresponding scale parameters for maintenance and deorbiting, respectively.

Because the environment evolves in discrete time, the continuous Weibull laws are converted into one-step transition probabilities. Following the hazard-based formulation implemented in the simulator, the probability that satellite s enters a maintenance state during the interval $[t, t + 1)$ is given by

$$p_s^m(o_{s,t}) = 1 - \exp \left\{ - \left[\left(\frac{o_{s,t} + 1}{\eta_s^m} \right)^{\beta_s^m} - \left(\frac{o_{s,t}}{\eta_s^m} \right)^{\beta_s^m} \right] \right\}, \quad (5)$$

while the probability of entering a deorbiting state over the same interval is

$$p_s^d(o_{s,t}) = 1 - \exp \left\{ - \left[\left(\frac{o_{s,t} + 1}{\eta_s^d} \right)^{\beta_s^d} - \left(\frac{o_{s,t}}{\eta_s^d} \right)^{\beta_s^d} \right] \right\}. \quad (6)$$

At each time step, a random variable $u_{s,t} \sim \mathcal{U}(0, 1)$ is drawn for each non-serviced satellite. The satellite state is then updated according to the following decision rule:

$$\text{if } u_{s,t} < p_s^d(o_{s,t}), \quad \text{satellite } s \text{ enters the deorbiting state,} \quad (7)$$

$$\text{else if } u_{s,t} < p_s^d(o_{s,t}) + p_s^m(o_{s,t}), \quad \text{satellite } s \text{ enters the maintenance state.} \quad (8)$$

This formulation gives priority to the deorbiting state when both risks are significant, reflecting the more critical nature of terminal failure. In addition, once a deorbiting need is triggered, it overrides any previously active maintenance or refuelling need in the satellite state representation, as it corresponds to an irreparable critical failure. Note also that service needs cannot overlap: a satellite holds at most one active service need at any given time. A satellite struck by a deorbiting failure is considered dead in place: it stops generating revenue and its degradation process is frozen until the servicer performs the disposal operation.

The Weibull shape parameters β_s^m and β_s^d determine the aging profile of the satellite. In particular, $\beta > 1$ corresponds to an increasing failure rate with age, $\beta = 1$ corresponds to a constant failure rate, and $\beta < 1$ corresponds to early-life failures. The scale parameters η_s^m and η_s^d determine the characteristic operating time at which the corresponding service need becomes significant. In the implementation, all time-related satellite parameters are converted into simulation steps so that the failure processes remain consistent with the selected time discretization.

Finally, each satellite of the fleet is associated with an annual revenue r_s , expressed in millions of dollars, which represents the income that the satellite generates for its operator while in nominal operation. Whenever a satellite has an active service need, i.e., refuelling, maintenance, or deorbiting, it is considered non-operational and stops generating revenue until the corresponding service is performed.

The outcome of a servicing operation depends on its type. Once a refuelling or maintenance operation is successfully completed, the satellite is considered “as good as new”: it resumes nominal operation and is removed from further degradation and failure generation for the remainder of the episode (one-shot service model). A deorbiting operation, in contrast, corresponds to the disposal of an irreversibly lost asset: upon completion of the service, the defective satellite is removed from

orbit and replaced by a brand-new satellite occupying the same orbital slot. This replacement incurs a cost equal to the asset value of the lost satellite, which is accounted for in the reward function (Section 3.2.3). For simplicity, the manufacturing, launch, and orbit insertion of the replacement satellite are assumed to be instantaneous: the new satellite enters service as soon as the deorbiting operation is completed, with its operating time reset to zero, and removed from further degradation and failure generation.

3.1.4 Service Vehicle

The service vehicle, denoted by v , moves through the orbital environment by executing the rendezvous transfers associated with arcs a (Section 3.1.5). At each decision epoch, it may initiate a transfer toward a client satellite or a refuelling depot, or simply wait on its current orbit, in which case it drifts naturally for one time step at no propellant cost (further details on the action model are given in Section 3.2.2). At the initial time step, $t = 0$, the vehicle is positioned at a predefined orbital position $(o_{v,0}, \theta_{v,0})$, defined by its starting orbit $o_{v,0} \in \mathcal{O}$ and its starting true anomaly $\theta_{v,0} \in [0^\circ, 360^\circ]$.

As discussed above, the servicer may operate under either a high-thrust or a low-thrust propulsion architecture. This propulsion mode is selected prior to training and remains unchanged throughout the mission. Because each orbital manoeuvre requires propellant, we denote by f_t the remaining onboard propellant mass, in kilograms, at time step t . Depending on the propulsion architecture considered, f_t corresponds to either the high-thrust or low-thrust fuel available to the servicer. Let F_{HT} and F_{LT} denote the maximum onboard propellant capacities of the servicer for the high-thrust and low-thrust configurations, respectively, both expressed in kilograms.

3.1.5 Arcs

The servicer moves through the orbital environment by executing transfers referred to as *arcs*. An arc represents one or several orbital manoeuvres that bring the servicer from its current orbital position to the current position of a target entity. Every object in the environment (servicer, client satellites, and refuelling depots) occupies a continuous orbital position, defined by the pair (o, θ) , where $o \in \mathcal{O}$ is the orbit and $\theta \in [0^\circ, 360^\circ]$ is the true anomaly on that orbit.

At each decision epoch t , a candidate arc $a_j(t)$ is constructed *on the fly* from the servicer's current position $(o_v, \theta_{v,t})$ to the current position $(o_j, \theta_{j,t})$ of each admissible target $j \in \mathcal{S} \cup \mathcal{D}$, where $\mathcal{S}$ is the set of client satellites and $\mathcal{D}$ the set of refuelling depots. Because all targets drift continuously along their orbits, the set of candidate arcs (and the parameters of each arc) are time-dependent: the same target reached at two different epochs generally yields different transfer geometries, durations, and propellant costs. Each arc is therefore a genuine *rendezvous* transfer: its duration includes any phasing or coasting required for the servicer to arrive co-located with the moving target.

We distinguish between two classes of arcs, corresponding to the high-thrust and low-thrust propulsion architectures. In this study, four types of high-thrust manoeuvres and two types of low-thrust manoeuvres are considered for connecting the servicer to a target.

For high-thrust arcs:

- *Phasing manoeuvre*: when the servicer and the target lie on the same orbit, they share the same orbital period and the true-anomaly gap $\Delta\theta$ between them never closes by natural drift. The gap is closed by an impulsive out-and-back phasing manoeuvre on a k -revolution phasing ellipse. The period of the phasing ellipse is

$$T_{\text{ph}} = \frac{2\pi k + \Delta\theta}{k n}, \quad (9)$$

where n is the mean motion of the orbit. The semi-major axis of the phasing ellipse follows from Kepler's third law, $a_{\text{ph}} = \left(\mu T_{\text{ph}}^2 / 4\pi^2\right)^{1/3}$, and the total impulse cost (entry and exit burns) is obtained from the vis-viva equation evaluated at the orbit radius r :

$$\Delta v_{\text{ph}} = 2 \left| \sqrt{\frac{2\mu}{r} - \frac{\mu}{a_{\text{ph}}}} - \sqrt{\frac{\mu}{r}} \right|, \quad \tau = k T_{\text{ph}}. \quad (10)$$

The number of phasing revolutions k is a model parameter that trades transfer time against propellant ($k = 18$ in the experiments of Chapter 4).

- *Hohmann transfer*: when the servicer and the target lie on coplanar orbits of different semi-major axes, the transfer is a classical two-burn Hohmann manoeuvre. A Hohmann transfer sweeps exactly 180° of true anomaly, so the servicer must depart at the correct relative phase

to intercept the drifting target. Rather than assuming this geometry, the arc includes a free coasting wait $t_w \geq 0$ before the first burn, obtained from the rendezvous condition

$$\theta_v + n_1 t_w + \pi \equiv \theta_j + n_2 (t_w + t_H) \pmod{2\pi}, \quad (11)$$

where n_1 and n_2 are the mean motions of the departure and arrival orbits and t_H is the Hohmann transfer time. Since a change of semi-major axis implies $n_1 \neq n_2$, the smallest non-negative solution always exists within one synodic period $T_{\text{syn}} = 2\pi/|n_1 - n_2|$. The waiting time carries no propellant cost but is included in the arc travel time, $\tau = t_w + t_H$, and therefore contributes directly to the mission scheduling cost.

- *Hohmann transfer with inclination change*: when the target orbit differs in both semi-major axis and inclination, the plane rotation of magnitude Δi is combined with one of the two Hohmann burns through the law of cosines, $\Delta v = \sqrt{v_a^2 + v_b^2 - 2v_a v_b \cos \Delta i}$. The combined plane change is executed either at departure or at arrival, whichever yields the lower total ΔV (in practice, at the apsis with the lower orbital velocity). The same rendezvous waiting logic as in the pure Hohmann case applies.
- *Plane change*: when the target orbit shares the same semi-major axis but a different inclination, the transfer is a single impulsive plane rotation performed at the line of nodes, with cost

$$\Delta v_{\text{pc}} = 2 v \sin\left(\frac{\Delta i}{2}\right), \quad (12)$$

where v is the circular orbital velocity. Because both orbits share the same period, any residual true-anomaly gap does not close by drifting; when the servicer and the target are not phase-aligned, a phasing manoeuvre of the form (9)–(10) is appended to the plane change, and both ΔV contributions are paid.

For low-thrust arcs:

- *Low-thrust phasing manoeuvre*: when the servicer and the target lie on the same orbit, the constant true-anomaly gap $\Delta\theta$ is closed using a bang–coast–bang continuous-thrust profile:

the servicer thrusts during $(1 - \alpha) \Delta t/2$, coasts during $\alpha \Delta t$, and thrusts again during $(1 - \alpha) \Delta t/2$, where Δt is the total manoeuvre duration and $\alpha \in [0, 1)$ is the coasting fraction ($\alpha = 0.5$ in this work). With a thrust acceleration $a_{\text{th}} = 2\eta P/(mc)$, where P is the available electric power, η the thruster efficiency, m the servicer mass, and $c = I_{\text{sp}} g_0$ the effective exhaust velocity, the phase displacement and impulse cost are

$$\Delta\theta = \frac{3}{4} \frac{a_{\text{th}} (1 - \alpha^2) \Delta t^2}{r_0}, \quad \Delta v = \frac{4 r_0 \Delta\theta}{3 (1 + \alpha) \Delta t}, \quad (13)$$

where r_0 is the orbit radius. The minimum-duration manoeuvre (full available power) is computed first; the duration is then rounded up to an integer number of simulation steps and the ΔV is re-evaluated for the stretched duration, which corresponds to flying the profile at reduced thrust.

- *Edelbaum transfer*: any change of semi-major axis, inclination, or both is modeled as a continuous-thrust Edelbaum spiral with cost

$$\Delta v = \sqrt{v_1^2 + v_2^2 - 2v_1 v_2 \cos\left(\frac{\pi}{2} \Delta i\right)}, \quad (14)$$

where v_1 and v_2 are the circular velocities of the initial and final orbits and Δi is the total inclination change. The transfer duration is derived from the propulsion system itself: the required propellant mass is obtained from the Tsiolkovsky equation, and the burn time follows from the mass flow rate $\dot{m} = T/c$, with thrust $T = 2\eta P/c$. Terminal phase alignment with the drifting target is assumed to be absorbed within the many-revolution spiral at negligible additional cost, a standard simplification for low-thrust transfer-time estimation.

Natural drift along the current orbit is captured by a dedicated *wait* action of the decision model (Section 3.2.2), during which the servicer coasts in place for exactly one time step at zero propellant cost. This decouples the decision to loiter, e.g., to wait for a more favorable phasing geometry, from the transfer model itself.

For both propulsion architectures, each arc a is characterized by the following parameters:

- τ_a : time required to traverse arc a , in s;

- ψ_a : propellant consumption required to traverse arc a , expressed in ΔV ;
- ρ_a : fuel consumption to traverse arc a , in kg.

Accordingly, each candidate arc is characterized by the parameter set $(p_v, p_j, \tau_a, \psi_a, \rho_a)$, where $p_v = (o_v, \theta_{v,t})$ and $p_j = (o_j, \theta_{j,t})$ denote the servicer's and the target's current orbital positions at the decision epoch. All durations are rounded up to an integer number of simulation steps to guarantee feasibility within the discrete-time environment. The fuel consumption ρ_a is evaluated at decision time through the Tsiolkovsky equation because it depends on the current (wet) mass of the servicer at the moment the manoeuvre is initiated. For low-thrust transfers, an additional control parameter is introduced, namely the specific impulse I_{sp} . In this case, the agent does not only select the next target, but also chooses the propulsion operating point by selecting an I_{sp} value from a discrete set (uniformly spaced over a predefined range) fixed prior to training. This choice determines both the available thrust, $T = 2\eta P / (I_{sp} g_0)$, and the propellant expenditure of the manoeuvre, and therefore governs the speed–efficiency trade-off of each transfer.

Algorithms 1 and 2 summarize how a low-thrust and a high-thrust arc are constructed on the fly between the servicer and a selected target, and how the associated manoeuvre parameters are computed for each feasible orbital transfer.

3.1.6 refuelling Infrastructure

The environment further incorporates orbital refuelling depots, which enable the service vehicle to replenish its onboard propellant during the mission. Like every other entity in the environment, each depot occupies a continuous orbital position defined by the pair (o, θ) , where $o \in \mathcal{O}$ is its orbit and $\theta \in [0^\circ, 360^\circ]$ is its true anomaly on that orbit. Depots revolve around the Earth in the same way as the client satellites, and their positions are propagated at every simulation step. Each depot is assumed to hold an unlimited propellant supply over the mission horizon.

A refuelling operation is triggered by selecting a depot as the next transfer target, in the same way as a client satellite (Section 3.2.2). Upon completion of the corresponding rendezvous arc, the servicer arrives co-located with the (drifted) depot and its propellant mass is restored to its maximum capacity, denoted by F_{HT} or F_{LT} depending on the selected propulsion mode. The

Algorithm 1 On-the-fly construction of a low-thrust arc between the servicer and a target

Require: Servicer position $p_v = (o_v, \theta_v)$, target position $p_j = (o_j, \theta_j)$, time step duration Δt , servicer mass m , I_{sp} chosen by the RL agent, propulsion data (P, η)

Ensure: Arc a with manoeuvre type, travel time, ΔV , and fuel consumption

Extract the orbital and angular characteristics of p_v and p_j

Determine whether p_v and p_j lie on the same orbit (same semi-major axis and inclination)

if p_v and p_j are on the same orbit **then**

 Define a *Low-thrust phasing* arc

 Compute the minimum bang–coast–bang duration and ΔV from the phasing model (13)

 Round the duration up to an integer number of steps and re-evaluate ΔV at the corresponding reduced thrust

else

 Define an *Edelbaum transfer* arc

 Compute ΔV from the Edelbaum model (14), the propellant mass from the Tsiolkovsky equation, and the burn time from the mass flow rate $\dot{m} = T/c$

 Round the duration up to an integer number of steps

end if

Compute the fuel consumption ρ_a from the Tsiolkovsky equation with the current servicer mass m

return a

propellant transfer itself is assumed to be completed within the arrival time step, at no additional time or propellant cost, a reasonable simplification given the coarse time discretization adopted in this work (one step corresponds to two days in the case studies of Chapter 4). Once refueled, the servicer is immediately free to select its next target or to wait on the depot’s orbit.

3.2 Event-Driven Semi-Markov Decision Process Formulation

A natural framework for modeling the sequential decision-making problem described above is the Markov Decision Process (MDP). This mathematical formulation is widely adopted in the reinforcement learning literature and provides a principled structure in which an agent, the decision maker, interacts with an environment by selecting actions at discrete time steps. In response, the environment returns a numerical reward signal that the agent seeks to maximise over time through its sequence of decisions. Formally, an MDP is characterized by the following elements: $\mathbb{S}$ denotes the set of possible environment states, $\mathbb{A}$ denotes the set of actions available to the agent, and $R(s, a)$ is the reward function that returns the immediate reward received by the agent after taking action a in state s .

Algorithm 2 On-the-fly construction of a high-thrust arc between the servicer and a target

Require: Servicer position $p_v = (o_v, \theta_v)$, target position $p_j = (o_j, \theta_j)$, time step duration Δt , servicer mass m , propulsion data (I_{sp})

Ensure: Arc a with manoeuvre type, travel time, ΔV , and fuel consumption

Extract the orbital and angular characteristics of p_v and p_j

Determine whether a semi-major axis change and/or an inclination change is required

if no semi-major axis change and no inclination change **then**

 Define a *Phasing* arc

 Compute ΔV and the duration $k T_{ph}$ from the k -revolution phasing-ellipse model (9)–(10)

else if semi-major axis change only **then**

 Define a *Hohmann transfer* arc

 Compute the rendezvous coasting wait t_w from (11) so that the 180° transfer intercepts the drifting target

 Compute ΔV from the two Hohmann burns and set $\tau = t_w + t_H$

else if semi-major axis change and inclination change **then**

 Define a *Hohmann transfer with inclination change* arc

 Combine the plane rotation with the departure or arrival burn, whichever minimizes the total

ΔV

 Include the rendezvous coasting wait t_w as in the pure Hohmann case

else ▷ inclination change only

 Define a *Plane change* arc with $\Delta V = 2v \sin(\Delta i/2)$

 If the true-anomaly gap is non-zero, append a phasing manoeuvre and add its ΔV and duration

end if

Round the duration up to an integer number of steps

Compute the fuel consumption ρ_a from the Tsiolkovsky equation with the current servicer mass m

return a

An important structural feature of the on-orbit servicing problem is that a single decision typically spans several simulation time steps: once a transfer is initiated, the servicer remains committed to it for the whole transit duration, and a servicing operation similarly occupies the vehicle for several steps during which no alternative action can be taken. The problem is therefore cast as an event-driven *semi*-Markov decision process: the agent acts only at decision epochs $k = 0, 1, \dots, K$, occurring at times t_k at which the servicer is idle and faces a genuine choice. Between two consecutive decisions, the environment is propagated autonomously one time step at a time, along the transit, the servicing operation at the destination, and any intermediate step in which the agent has no real influence. The per-step rewards collected along the way are then summed up into a single decision-level reward,

$$R_k = \sum_{t=t_k+1}^{t_{k+1}} r_t. \quad (15)$$

This event-driven formulation substantially shortens the effective decision horizon seen by the agent and improves credit assignment over long missions, since the discount factor is applied between successive decisions rather than between raw simulation steps. The following subsections describe the state, action, and reward components of this formulation in the context of the proposed on-orbit servicing problem.

3.2.1 State Definition

At each decision epoch, the agent observes a state $s_t \in \mathbb{S}$ that must contain all the information required to support effective decision-making. In the proposed formulation, the observation is organized into four groups of features: (i) the servicer state, (ii) per-satellite features, (iii) per-depot features, and (iv) global fleet-level aggregates. All continuous features are normalized by the mission horizon T , the propellant capacity, or the angular range 360° , so that every component lies in $[0, 1]$ (or $[-1, 1]$ for signed quantities), which improves training stability. Table 3.1 summarizes the complete observation, whose total dimension is $14|\mathcal{S}| + 3|\mathcal{D}| + 17$.

The *servicer features* describe the vehicle’s own situation: its current orbit and true anomaly, its remaining propellant mass and a low-fuel warning flag, the current mission time, and its operational status, namely whether it is in transit or performing a servicing operation, together with the

remaining duration and, for servicing, a one-hot encoding of the ongoing operation type.

The *per-satellite features* combine raw state variables with engineered quantities that expose the underlying service-demand dynamics to the agent. In addition to each satellite’s true anomaly, serviced flag, and active need indicators (refuelling, maintenance, or deorbiting, with the elapsed time since each need was triggered), the observation includes two Weibull pressure ratios, $o_{s,t}/\eta_s^m$ and $o_{s,t}/\eta_s^d$, which act as direct proxies for the instantaneous failure intensity of each satellite and spare the policy from having to infer the (β, η) parameters implicitly. It also includes the signed margin remaining before the deterministic fuel failure threshold. The local transfer geometry is exposed directly in the observation: for every satellite, the propellant cost and travel time of the cheapest single arc from the servicer’s current position (minimized over the available I_{sp} settings in the low-thrust case) are computed on the fly and included as features, with unreachable targets mapped to a saturated sentinel value. A binary flag also indicates whether each satellite lies on the servicer’s current orbit, distinguishing inexpensive intra-orbit phasing opportunities from costly plane or altitude changes.

The *per-depot features* mirror this structure for the refuelling infrastructure: the true anomaly of each depot and the propellant cost and travel time required to reach it. Finally, the *global aggregates* provide the fleet-level fractions of satellites with pending refuelling, maintenance, and deorbiting needs, the fraction of satellites not yet serviced, and the fraction of satellites currently non-operational. These inexpensive scalars summarize information that the value function would otherwise have to recompute from the per-satellite channels.

3.2.2 Action Definition

As discussed previously, at each decision epoch the servicer selects the next target to visit, i.e., a client satellite or a refuelling depot, or chooses to wait, in which case it coasts in place and drifts naturally along its current orbit for one time step at no propellant cost. Selecting a target triggers the on-the-fly construction of the corresponding rendezvous arc from the servicer’s current position, as described in Section 3.1.5. In the low-thrust configuration, the action additionally specifies the propulsion operating point: each action is a (target, I_{sp}) pair, where the specific impulse is chosen

Table 3.1: Components of the observation space. $|\mathcal{S}|$ and $|\mathcal{D}|$ denote the numbers of client satellites and refuelling depots, respectively.

Component	Dim.	Description
<i>Servicer</i>		
Current orbit	1	Index of the servicer’s current orbit $o_v \in \mathcal{O}$
Current true anomaly	1	$\theta_v/360^\circ$
Remaining propellant	1	f_t normalized by F_{HT} or F_{LT}
Low-fuel flag	1	1 if the fuel ratio is below a warning threshold
Mission time	1	t/T
In-transit flag	1	1 while executing a transfer arc
Remaining transit time	1	Remaining steps of the ongoing transfer, $/T$
In-servicing flag	1	1 while performing a servicing operation
Remaining service time	1	Remaining steps of the ongoing service, $/T$
Service type	3	One-hot encoding (R, M, D) of the ongoing operation
<i>Client satellites (per satellite)</i>		
Serviced flag	$ \mathcal{S} $	1 if already serviced (R or M completed)
True anomaly	$ \mathcal{S} $	$\theta_s/360^\circ$
Same-orbit flag	$ \mathcal{S} $	1 if on the servicer’s current orbit
Active-need flags	$3 \mathcal{S} $	Binary indicators of pending R, M, D needs
Need ages	$3 \mathcal{S} $	Time since each need activation, $/T$
Weibull pressure	$2 \mathcal{S} $	$o_{s,t}/\eta_s^m$ and $o_{s,t}/\eta_s^d$ (capped)
Fuel-failure margin	$ \mathcal{S} $	$(\bar{o}_s^r - o_{s,t})/T$, clipped to $[-1, 1]$
Fuel cost to satellite	$ \mathcal{S} $	Cheapest single-arc ρ_a , normalized by capacity
Travel time to satellite	$ \mathcal{S} $	Corresponding τ_a in steps, $/T$
<i>refuelling depots (per depot)</i>		
True anomaly	$ \mathcal{D} $	$\theta_d/360^\circ$
Fuel cost to depot	$ \mathcal{D} $	Cheapest single-arc ρ_a , normalized by capacity
Travel time to depot	$ \mathcal{D} $	Corresponding τ_a in steps, $/T$
<i>Global aggregates</i>		
Pending-need fractions	3	Fleet fractions with pending R, M, D needs
Unserviced fraction	1	Fraction of satellites not yet serviced
Non-operational fraction	1	Fraction of satellites currently failing

from the discrete set defined prior to training. The action space is therefore of size

$$|\mathbb{A}| = (|\mathcal{S}| + |\mathcal{D}|) n_{isp} + 1, \quad (16)$$

where n_{isp} denotes the number of selectable I_{sp} values ($n_{isp} = 1$ in the high-thrust configuration) and the additional action corresponds to the wait decision. Note that any unserviced satellite may be targeted, whether or not it has an active service need, which enables preemptive interventions such as refuelling within the anticipation window H introduced in Section 3.1.3.

However, not all actions are feasible at every decision epoch. An action is excluded from the admissible set whenever (i) the target is a satellite that has already been serviced, (ii) the target is co-located with the servicer, making the transfer degenerate, or (iii) the propellant required by the corresponding arc exceeds the servicer’s remaining fuel, $\rho_a > f_t$. The wait action, by contrast, is always admissible. Consequently, the set of admissible actions depends on the current state of the environment, and in particular on the servicer’s fuel level and position. Notably, when no target remains both reachable and affordable, only the wait action is left and the servicer is considered stranded: the episode then terminates as an out-of-fuel failure if there are still some satellites left to visit. This design choice makes propellant a genuinely binding resource (an empty tank immobilizes the vehicle) and is what renders the refuelling decisions economically meaningful.

To handle these state-dependent feasibility constraints, invalid actions are filtered using an action mask. The detailed implementation of this masking procedure is presented in Section 3.4.2.

3.2.3 Reward Definition

The reward function is a critical component of any reinforcement learning model: it encodes the designer’s objectives into a scalar signal and thereby guides the agent’s behaviour. The reward function adopts an integrated-system net present value (NPV) system: the servicer and the satellite operators are modeled as a single economic entity. What the agent is rewarded for is the operational revenue preserved across the fleet, net of irreversible asset write-offs and propellant expenditure.

Let $\bar{r}_s = r_s \Delta t / T_{yr}$ denote the revenue accrued by satellite s during one simulation step while operational, with r_s its annual revenue and $T_{yr} = 365.25$ days, and let Π_s denote its replacement

(asset) value. We further define the operational indicator $y_{s,t} \in \{0, 1\}$, equal to 1 if satellite s is operational at step t , i.e., it has no pending service need, is not currently under service, and is not dead in place awaiting deorbiting. Already-serviced satellites remain operational until the end of the horizon under the one-shot service model of Section 3.1.3.

At each simulation step t , the immediate reward is decomposed into four additive components:

$$r_t = r_t^{\text{rev}} + r_t^{\text{risk}} + r_t^{\text{fuel}} + r_t^{\text{oof}}. \quad (17)$$

r_t^{rev} is the *forgone-revenue term*. Rather than paying the gross revenue $\sum_s \bar{r}_s y_{s,t}$ of the operational satellites, the reward is centered on the healthy baseline: a fully operational fleet earns 0, and every non-earning satellite costs its per-step revenue,

$$r_t^{\text{rev}} = - \sum_{s \in \mathcal{S}} \bar{r}_s (1 - y_{s,t}). \quad (18)$$

The two framings differ only by the constant $\sum_s \bar{r}_s$ per step and therefore induce the same optimal policy; the centered form, however, removes the large, near-constant passive income that would otherwise dominate the return and drown the advantage signal during policy optimisation. This term generalizes the delay penalty of classical formulations: the economic cost of an unmet service need is exactly the revenue forgone while the satellite is out of service, and it applies uniformly to all outage causes, i.e., pending refuelling or maintenance needs, satellites dead in place awaiting disposal, and satellites immobilized during a servicing operation.

r_t^{risk} is the *expected asset-loss term*. A deorbiting failure irreversibly destroys a satellite of value Π_s . Booking this loss as a lumpy penalty $-\Pi_s$ at the (rare) failure event would inject into the return sparse spikes worth several hundred times a step's revenue, which the value function cannot fit reliably. The loss is therefore charged in expectation. Indeed, at every step, each unserviced, operational, and still-aging satellite contributes its instantaneous deorbit hazard times its asset value,

$$r_t^{\text{risk}} = - \sum_{s \in \mathcal{S}_t^{\text{alive}}} p_s^d(o_{s,t}) \Pi_s, \quad (19)$$

where $\mathcal{S}_t^{\text{alive}}$ denotes this set of at-risk satellites and p_s^d is the one-step deorbit probability of Eq. 6. This smooth surrogate has the same expectation as the lumpy write-off but a per-step magnitude on the order of one step's revenue, yielding a low-variance learning signal. It also makes the benefit of preventive servicing directly visible to the agent: servicing a satellite halts its aging and thus extinguishes its accruing risk charge. The realized asset losses are tracked separately for the economic reporting of Section 3.3.

r_t^{fuel} is the *propellant cost*. Each orbital manoeuvre incurs a penalty proportional to the propellant consumed, applied once upon completion of the transfer:

$$r_t^{\text{fuel}} = \begin{cases} -\rho_a c^{\text{fuel}}, & \text{if the servicer completes arc } a \text{ at step } t, \\ 0, & \text{otherwise,} \end{cases} \quad (20)$$

where ρ_a is the propellant mass consumed along arc a (in kg) and c^{fuel} is the propellant unit price in \$/kg, which differs between the high-thrust and low-thrust configurations (Section 4.1.5).

r_t^{eof} is the *out-of-fuel failure penalty*. Because the action mask of Section 3.4.2 already prevents the agent from selecting transfers it cannot afford, the invalid-action penalty is not necessary. The residual failure mode is being *stranded*: if, at some decision epoch, no target remains both reachable and affordable, the mission is irrecoverably lost. The episode then terminates with a penalty equal to the fleet revenue the servicer can no longer protect over the remaining horizon,

$$r_t^{\text{eof}} = \begin{cases} -(T-t) \sum_{s \in \mathcal{S}} \bar{r}_s, & \text{if the servicer is stranded at step } t, \\ 0, & \text{otherwise.} \end{cases} \quad (21)$$

This penalty is deliberately sized so that early termination can never be an escape from the accruing forgone-revenue charges of Eq. (18), while a deliberate loiter, i.e., waiting while targets remain affordable, stays free of charge.

Summed over the horizon, the episode return

$$R = \sum_{t=0}^{T-1} r_t \quad (22)$$

coincides, up to the constant centering offset $T \sum_s \bar{r}_s$, with the realized fleet NPV: the total revenue earned by operational satellites, minus the value of irreversibly lost assets and the total propellant cost. Maximizing the expected return therefore directly maximises the expected economic value of the servicing campaign.

3.3 Economic Model

The reward function of Section 3.2.3 encodes, step by step, the economic consequences of the agent’s decisions. This section formalizes the full economic evaluation model built on top of it, which is used throughout Chapter 4 to compare servicing architectures, depot infrastructures, and scheduling policies on a single monetary scale: the cash flows entering the balance, the discounting and cost-attribution conventions used to aggregate them into a net present value (NPV), and the exact relation between the NPV and the training reward.

3.3.1 Cash Flows and Net Present Value

On the income side, each client satellite s generates the per-step revenue $\bar{r}_s$ introduced in Section 3.2.3 whenever its operational indicator $y_{s,t}$ equals one. On the expense side, the model accounts for four cost streams: (i) the initial investment I_0 , which combines the manufacturing and launch cost of the servicer with the share of the refuelling-depot price charged to the mission (Section 3.3.2); (ii) the depot station-keeping cost, a fixed daily outlay c_{sk} per depot; (iii) the propellant expenditure, each transfer arc a consuming a mass ρ_a of propellant priced at c^{fuel} per kilogram; and (iv) the asset write-offs, incurred whenever a satellite is irreversibly lost to a deorbiting failure or replaced by a fresh spacecraft.

Because money available today can be reinvested and is therefore worth more than the same amount received in the future, cash flows occurring at different epochs are not directly commensurable. Every cash flow is therefore discounted back to the start of the mission at an annual discount rate δ , through the discount factor

$$D(t) = (1 + \delta)^{-t \Delta t / T_{\text{yr}}}, \quad (23)$$

where $t \Delta t$ is the elapsed mission time at step t . We adopt $\delta = 8\%$ throughout this work, a value representative of the discount rates commonly used in economic assessments of space systems; setting $\delta = 0$ recovers simple undiscounted sums. Denoting by CF_t the net operating cash flow of step t — revenue of the operational satellites, minus propellant expenditure, station-keeping outlays, and asset write-offs booked at that step — the mission NPV reads

$$\text{NPV} = -I_0 + \sum_{t=1}^T D(t) CF_t, \quad (24)$$

the initial investment being a one-off outlay incurred at $t = 0$ and therefore entering the balance undiscounted.

3.3.2 Infrastructure Costs and Capital-Expenditure Proration

Two conventions ensure that capital expenditures are charged in proportion to the fraction of each asset's life actually consumed within the study window, rather than written off in full against missions that use only part of it.

First, the refuelling depot outlives every mission horizon considered in this work. Its acquisition price Π_{depot} is therefore spread over its design life L_{depot} , taken as 15 years, and each mission is charged only the share corresponding to its own duration,

$$I_0^{\text{depot}} = \Pi_{\text{depot}} \min\left(\frac{T \Delta t}{L_{\text{depot}}}, 1\right) \times n_{\text{depots}}. \quad (25)$$

The daily station-keeping cost c_{sk} , in contrast, is an operating expense: it is booked day by day over the horizon and discounted through Eq. (23) like any other recurring flow. Both terms are independent of the scheduling decisions, a property exploited in Section 3.3.3; their numerical values are specified with the study-case parameters of Chapter 4.

Second, the cost booked for a replaced satellite — whether a deorbit-induced asset loss under a servicing policy or a routine replacement in the no-service baseline used in Chapter 4 — is prorated

by the remaining mission time relative to the satellite's nominal design lifetime:

$$C_{\text{repl}}(t) = \Pi_s \frac{T - t}{L_{\text{nom}}}, \quad (26)$$

where Π_s is the satellite asset value, L_{nom} its nominal design lifetime (≈ 12 years, Section 3.1.3), $T - t$ and L_{nom} are expressed in the same time unit, and the charge is capped at Π_s . Writing off the full asset value would be unrealistic when only a small fraction of the satellite's life is at stake within the mission: losing a satellite when only a few months of mission remain should not be charged the same as losing a full-lifetime asset.

3.3.3 Relation to the Reward Function

The reward of Section 3.2.3 is precisely the action-dependent, undiscounted share of the NPV defined above. The infrastructure costs of Section 3.3.2, i.e., the initial investment and the depot station-keeping outlays, do not depend on the agent's decisions: being policy-independent, they shift the return of every policy by the same constant and leave the optimal policy unchanged. They are therefore omitted from the training signal and reinstated in the ex-post economic accounting of Chapter 4. Two further accounting differences are deliberate. First, the reward charges deorbiting risk in "expectation" through the hazard term of Eq. (19), whereas the NPV books the "realized" losses at their prorated value of Eq. (26); the two coincide in expectation, but the former provides a far lower-variance learning signal. Second, the return of Eq. (22) is undiscounted, while the reported NPV applies the financial discounting of Eq. (23) ex post to the realized cash flows.

3.4 Reinforcement Learning framework

The objective of the reinforcement learning agent is to find a policy π^* that maximises the expected cumulative return over the mission horizon:

$$\pi^* = \arg \max_{\pi} \mathbb{E}_{\pi} \left[\sum_{t=0}^{T-1} r_t \right]. \quad (27)$$

A key advantage of reinforcement learning over static optimisation methods lies in its inherently

adaptive decision-making process. Whereas static approaches compute the entire servicing plan in advance based on fixed input data, a reinforcement learning agent makes decisions dynamically: at each time step, it receives an updated observation of the environment state and adjusts its actions accordingly to maximise long-term cumulative reward. This online decision-making capability allows the agent to naturally accommodate the stochastic elements of the problem, such as the failure and degradation processes described in Section 3.1.3, by continuously adapting its servicing trajectory in response to the evolving operational conditions, thereby improving overall mission efficiency.

3.4.1 Algorithm Selection

The policy is optimized using Proximal Policy optimisation (PPO) (Schulman, Wolski, Dhariwal, Radford, & Klimov, 2017), a policy-gradient algorithm that has become a standard baseline in deep reinforcement learning owing to its stability and sample efficiency. PPO constrains each policy update by clipping the probability ratio between the new and old policies, thereby preventing destructively large updates while still allowing meaningful learning steps. The surrogate objective maximised at each iteration is

$$L^{\text{CLIP}}(\theta) = \hat{\mathbb{E}}_t \left[\min \left(\varrho_t(\theta) \hat{A}_t, \text{clip}(\varrho_t(\theta), 1 - \epsilon, 1 + \epsilon) \hat{A}_t \right) \right], \quad (28)$$

where $\varrho_t(\theta) = \pi_\theta(a_t \mid s_t) / \pi_{\theta_{\text{old}}}(a_t \mid s_t)$ is the importance sampling ratio, $\hat{A}_t$ is the estimated advantage, and ϵ is the clipping parameter.

3.4.2 Invalid Action Masking

Because only a subset of actions is feasible in any given state (Section 3.2.2), the standard PPO formulation must be adapted to handle the state dependent action spaces. We employ the invalid action masking technique (Huang & Ontañón, 2022), which modifies the policy network output prior to action sampling. Specifically, let $\mathbf{z}_t \in \mathbb{R}^{|\mathbb{A}|}$ denote the raw logits produced by the policy network at decision epoch t , and let $\mathcal{M}_t \subseteq \mathbb{A}$ be the set of valid actions determined by the action

mask. The masked policy is obtained by setting

$$z_{t,a} \leftarrow \begin{cases} z_{t,a}, & \text{if } a \in \mathcal{M}_t, \\ -\infty, & \text{otherwise,} \end{cases} \quad (29)$$

before applying the softmax operator to produce the action probability distribution $\pi_\theta(a \mid s_t)$. This ensures that infeasible actions receive zero probability, eliminating wasted exploration on actions that are known in advance to be invalid and thereby significantly improving training efficiency.

The action mask $\mathcal{M}_t$ is constructed at each decision epoch by evaluating, on the fly, the rendezvous arc from the servicer’s current position to the current position of every candidate target (Section 3.1.5). A (target, I_{sp}) action is retained in $\mathcal{M}_t$ if and only if the corresponding arc is structurally feasible, the target is neither an already-serviced satellite nor co-located with the servicer, and the propellant requirement of the arc does not exceed the servicer’s current fuel level, $\rho_a \leq f_t$. Note that in the low-thrust configuration affordability is assessed per operating point: for a given target, fuel-intensive I_{sp} settings may be masked while more efficient ones remain admissible. The wait action is always valid, which guarantees that $\mathcal{M}_t$ is never empty, as required by the maskable PPO implementation; when the wait action is the only one remaining, the servicer is stranded and the episode terminates with the out-of-fuel penalty of Eq. (21). Finally, the arc evaluations performed for the mask are shared with the per-target transfer-cost features of the observation (Section 3.2.1), so that feasibility checking incurs no additional computational overhead.

3.5 Scheduling Policies

The case studies of Chapter 4 evaluate two scheduling policies on this environment: the reinforcement-learning agent trained with the Maskable PPO algorithm of Section 3.4, and a greedy heuristic. Both operate through the identical interface: they receive the same observations, are constrained by the same feasibility mask, and are evaluated on the same stochastic seeds. All of this ensures that any performance difference is attributable to the quality of their decisions alone. The heuristic constitutes a transparent, training-free benchmark grounded in the same economic quantities as the reward function; comparing the two tests whether horizon-level, anticipatory optimisation adds value over

informed myopic decision-making.

3.5.1 Greedy Value-at-Stake Heuristic

The greedy policy encodes the intuitive fleet-operator strategy: serve the most valuable need first, and refuel when running low. At each decision epoch, it applies three rules in order. First, if the servicer’s remaining propellant fraction falls below a predefined warning threshold specified for the considered case study, it heads to the reachable depot with the cheapest transfer, with ties broken by travel time. Otherwise, it considers every satellite that is not yet serviced and either has a pending service need or has entered the proactive refuelling window of Section 3.2.2, and targets the one with the largest *value-at-stake* score,

$$V_s(t) = \bar{r}_s \max((T - t) - \tau_s(t), 0) + P_s^d(t) \Pi_s, \quad (30)$$

where $\tau_s(t)$ is the time remaining before the satellite’s next scheduled refuelling deadline (zero if a need is already pending) and $P_s^d(t)$ is its probability of suffering a deorbiting failure over the remaining window, obtained from the Weibull hazard of Section 3.1.3. The first term is the revenue that would be forgone if the satellite were never serviced. The second is its expected asset loss: the same two quantities that drive the reward function of Section 3.2.3, evaluated here myopically, one satellite at a time. For each candidate target, the heuristic selects the cheapest feasible transfer option, ties broken in favor of faster arcs; if no serviceable satellite is affordable, it waits.

This heuristic is deterministic given the state, costs nothing to train, and is fully interpretable, which makes it a strong domain-informed baseline rather than a strawman. Its limitations are structural. It evaluates interventions one at a time and therefore cannot sequence targets, bundle geometrically close services, or anticipate the clustering of future demand. The transfer cost enters its decision only through feasibility and tie-breaking, never through the objective itself, so it will pay an expensive plane change for a marginally higher score. And its fuel-reserve threshold is a static warning level rather than a guaranteed-return margin: an expensive outbound transfer accepted just above the threshold can leave the servicer unable to reach any depot afterwards, i.e., stranded.

3.5.2 Reinforcement-Learning Policy

The second policy is the Maskable PPO agent presented in Section 3.4, and it addresses precisely these structural limitations. By maximizing the expected cumulative decision-level reward of Eq. (15) over the whole horizon, it learns to weigh the immediate cost of a transfer against the future revenue it protects, to anticipate upcoming demand, and to manage its propellant as a state-dependent strategic resource rather than through a fixed reserve. Because the transfer and failure dynamics are stochastic, the policy is recomputed at every decision epoch from the current state, which makes it natively adaptive to the realized failure history. The invalid-action mask of Section 3.4.2 guarantees a non-empty action set at every epoch. When the always-valid wait action is the only option remaining and if there are still satellites to service, the servicer is stranded and the episode terminates with the out-of-fuel penalty of Eq. (21). These capabilities come at the price of a training cost and of a policy that is less directly interpretable than the greedy rule; the network architecture and hyperparameters are reported in Appendix A.

3.6 Framework Overview

Figure 3.1 provides a simplified overview of the complete agent-environment interaction loop of the proposed framework, gathering together the components introduced throughout this section. The blue blocks denote the simulation environment, whereas the red blocks denote the Maskable PPO reinforcement-learning agent.

At each decision epoch, the agent emits an action that the environment first decodes. A "wait" action simply lets the servicer drift in place for a single time step, whereas a (target, I_{sp}) action triggers the on-the-fly construction of the corresponding rendezvous arc from the servicer's current position. The environment then advances the system one time step at a time, covering the transfer and, upon arrival, the servicing of the selected satellite or the refuelling of the servicer at a depot. At each step, the fleet state is updated consistently with the event driven semi Markov decision process introduced above. Once the next decision epoch is reached, the resulting observation and its associated reward are assembled. Finally, the action mask is computed from this new observation and applied to the output of the actor network, so that the agent draws its next action only from the

set of admissible actions, thereby closing the loop.

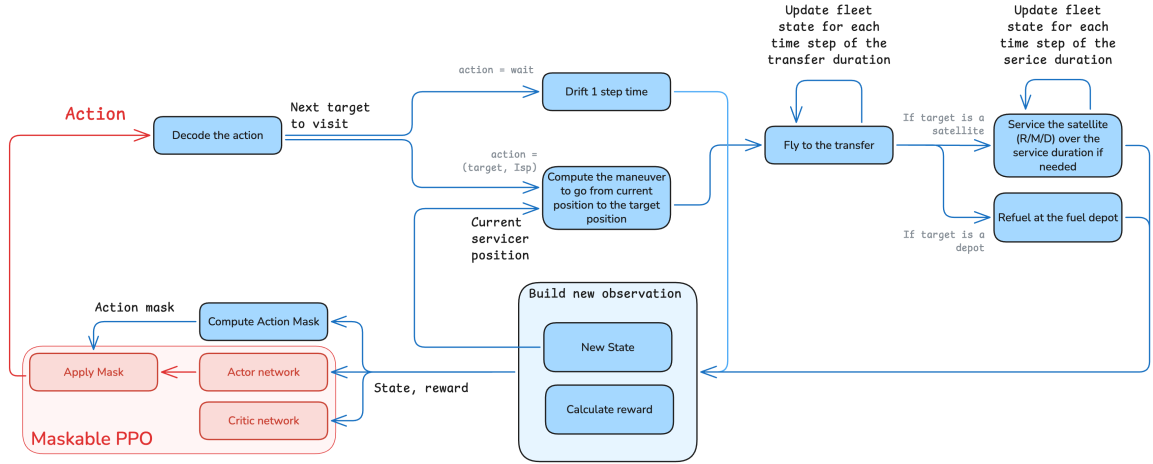


Figure 3.1: Pipeline of the proposed framework, illustrating one full decision cycle of the event-driven semi-Markov decision process. Blue blocks denote the simulation environment while red blocks denote the Maskable PPO agent.

Chapter 4

Results

In this Chapter, the proposed IOOSSO framework is applied to several case studies designed to evaluate its performance and to investigate the effects of service demand intensity and mission time horizon on the behaviour of both high-thrust and low-thrust propulsion modes.

4.1 Assumptions and Scenarios

This subsection first describes the orbital environment adopted for case studies, then introduces the scenarios considered, and finally details the assumptions and initial conditions defined for the satellite fleet, refuelling depots, and servicer spacecraft.

4.1.1 Orbital Environment

Table 4.1 summarizes the orbital network used across all first part of case studies. The network comprises three circular MEO orbits. Although the present study focuses exclusively on MEO configurations, the framework is general and can be applied to any orbital regime.

The three MEO orbits are inspired by those used in the SES o3b mPOWER satellite constellation. This orbital regime was selected because MEO satellites typically have long operational lifetimes, on the order of 10 to 15 years Liu et al. (2025); SES (2016), which provides a strong economic justification for the types of on-orbit servicing operations considered in this work. Figure 4.1 provides a three-dimensional visualization of the three MEO orbits described in table 4.1.

Orbit id	Period (sec)	Semi major axis (km)	Inclination (deg)	RAAN (deg)	Based on
1^M	17100	14345.8	0.0	0.0	o3b mPOWER
2^M	17600	14624.1	3.0	0.0	o3b mPOWER
3^M	17600	14624.1	0.0	0.0	o3b mPOWER

Table 4.1: Orbital parameters used for the first part of case study

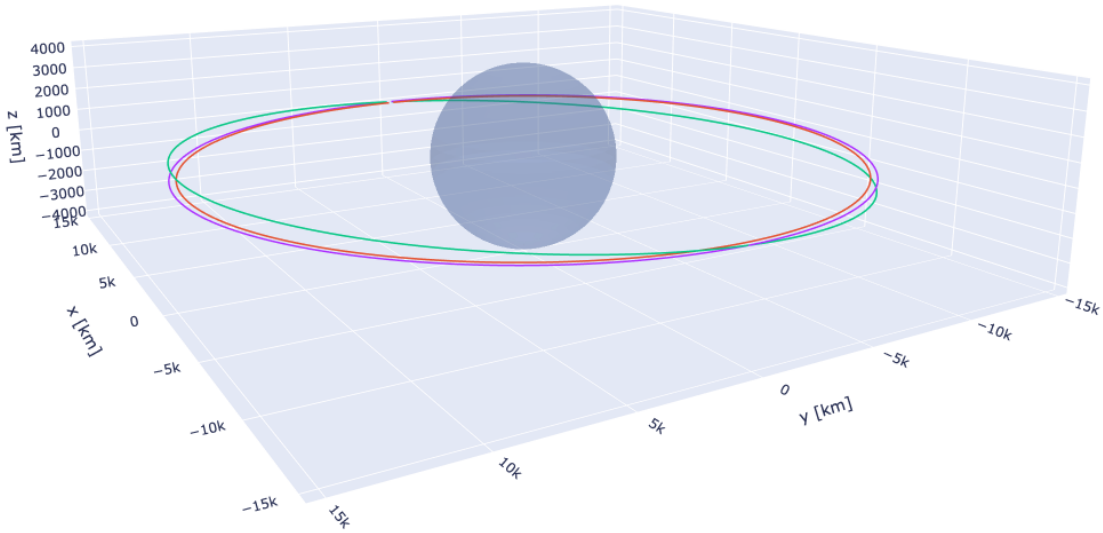


Figure 4.1: Visualization of the MEO orbits used

4.1.2 Case-Study Scenarios

Several case studies were designed to assess the influence of three key parameters on the behaviour of the high-thrust (HT) and low-thrust (LT) servicers: the mission time horizon, the size of the client-satellite fleet (that is, the level of service demand), and the number and position of refuelling depots. The scenarios are presented below in order of increasing complexity.

We first introduce, in Section 4.2.1, a deliberately simple scenario comprising a fleet of seven satellites distributed across the three MEO orbits introduced in Section 4.1.1, served by a single refuelling depot over a two-year horizon. For this scenario only, the Weibull parameters and the

initial operating times of the satellites are adjusted so that every satellite develops a service need early in the episode, making it easy to observe the servicer performing the full range of servicing operations. This case study is purely illustrative: its sole purpose is to show how the trained reinforcement-learning agent behaves and how its decisions translate into concrete servicing actions.

Section 4.2.2 then examines a case study combining a small client fleet (30 satellites), a single refuelling depot, and a short, one-year mission horizon. Building on this, Section 4.2.3 extends the horizon to five years for the same 30-satellite, single-depot configuration, allowing the short and long horizon results to be compared directly while all other parameters are held fixed.

Section 4.2.4 then isolates the effect of service demand by increasing the client-fleet size across 30, 50, 70, and 90 satellites, while keeping a single refuelling depot and the five-year horizon.

Section 4.2.5 varies the number of refuelling depots (one, two, and three) for each of these four fleet sizes, again over the five-year horizon, to assess the interplay between the refuelling infrastructure and the propulsion architecture. The influence of depot placement is then investigated through single-depot case studies in which the depot’s initial orbit is varied.

Finally, Section 4.2.6 compares the RL policies with the greedy policy introduced in Section 3.5.1, in order to assess the conditions under which reinforcement learning provides a meaningful scheduling advantage.

4.1.3 Customer Fleet Assumptions

Across all case studies, the client satellites are assumed to have a nominal design lifetime of 4,383 days (approximately 12 years), which also defines the deterministic fuel-failure threshold $\bar{o}_s^r$. To generate operationally realistic service demand from the very beginning of the mission, the fleet does not start in a brand-new condition: each satellite enters the simulation with a heterogeneous initial operating time, drawn between 2,000 and 4,300 days (roughly 5.5 to 11.7 years). The fleet therefore represents a mature, mid-to-late-life constellation in which refuelling deadlines and degradation-driven failures are imminent concerns rather than distant events — a prerequisite for meaningful scheduling decisions over the short, one-year mission horizons. Tables 4.2, 4.3, and 4.4 summarize the parameters adopted for the deterministic service-need model, the stochastic

service-need model, and the fleet economics, respectively.

Parameter	refuelling
Fuel failure threshold [days]	4,383
Service duration [days]	16
refuelling anticipation window H [days]	50

Table 4.2: Parameters of deterministic service needs

Parameter	Maintenance	Deorbitation
Service duration [days]	30	20
η [day]	5,600	6,100
β	3	4

Table 4.3: Parameters of stochastic service needs

Parameter	Value
Annual revenue r_s [\$M/year]	43
Replacement (asset) value Π_s [\$M]	150

Table 4.4: Economic parameters of the client satellites

The Weibull parameters governing the stochastic emergence of maintenance and deorbiting needs are chosen to reflect the expected aging behaviour of MEO satellites, whose design lifetimes are typically on the order of 10 to 15 years Liu et al. (2025), SES (2016). The scale parameters are deliberately set beyond the nominal design lifetime, i.e., $\eta_s^m = 5,600$ days for maintenance and $\eta_s^d = 6,100$ days for deorbiting, reflecting the fact that well-designed spacecraft are engineered to reach their design life with margin, so that the characteristic onset of degradation lies near and beyond the end of nominal operations. Under this calibration, the cumulative probability of having developed a maintenance need by the end of the 12-year design life is approximately 38%, while that of a critical deorbiting failure is approximately 23%; overall, roughly half of the fleet is expected to experience at least one stochastic service event within its design life, while the other half survives

to the deterministic refuelling deadline. This yields a balanced and operationally meaningful mix of predictable and unpredictable service demand.

The shape parameters are set to $\beta_s^m = 3$ for the maintenance process and $\beta_s^d = 4$ for the deorbiting process. A value of $\beta > 1$ produces an increasing hazard rate, meaning that the instantaneous probability of failure grows with operational age. This behaviour is characteristic of wear-out failure modes, in which cumulative mechanical degradation, thermal cycling, and radiation damage progressively reduce spacecraft reliability over time. The slightly lower shape parameter adopted for maintenance produces an earlier and more spread-out onset of repairable anomalies over the satellite’s life, whereas the steeper deorbiting profile concentrates critical, irreparable failures in the late portion of the lifetime. This is consistent with the intuition that terminal failures are rarer, and occur later, than repairable ones. In both cases the probability of early-life failure remains very low, in line with the high-reliability design standards of modern communication satellites. Although precise failure data for MEO satellites remain limited in the open literature, the selected parameters are representative of the general aging profiles reported in spacecraft reliability studies and are sufficient to generate operationally realistic service demand patterns for the purpose of evaluating the proposed scheduling framework.

The annual revenue r_s is derived from an estimate of the per-satellite revenue generated by the O3b and O3b mPOWER constellations operated by SES. Because detailed financial data at the individual satellite level are proprietary and not publicly disclosed, the per-satellite revenue is approximated by distributing the total constellation revenue, as reported in publicly available corporate filings, evenly across the number of operational satellites, yielding the value of approximately \$43M per satellite and per year reported in Table 4.4. This annual figure is then converted into the per-step revenue $\bar{r}_s$ used by the reward function of Section 3.2.3, and thus directly quantifies both the income earned by an operational satellite and the revenue forgone for each step during which a satellite with an active service need remains unserved. It should be noted that this approximation assumes a uniform revenue contribution across all satellites in the constellation, which is a simplification; in practice, individual satellite revenues may vary depending on coverage area, traffic demand, and contractual arrangements. Nevertheless, in the absence of publicly available per-asset

financial data (which remain extremely difficult to obtain for commercial satellite operators) this estimation provides a reasonable and internally consistent basis for quantifying the economic urgency of timely intervention within the proposed framework.

The replacement value Π_s represents the capital cost of restoring a lost orbital slot i.e., the manufacturing, launch, and orbit insertion of a substitute spacecraft, and is the quantity the asset value at stake in the expected-failure-cost term of the reward (Section 3.2.3) and in the prorated asset-loss accounting (Eq. 26). It is set to \$150M per satellite, a figure consistent with publicly reported procurement and launch cost estimates for MEO communication satellites of this class. Like the revenue estimate, it should be regarded as a representative order of magnitude rather than a precise per-asset valuation.

All servicing operations, whether deterministic (refuelling) or stochastic (maintenance and de-orbiting), are associated with service durations that depend on the type of intervention performed. These values are partly based on reference data reported in the open literature (Sarton Du Jonchay, Chen, Isaji, Shimane, and Ho (2022), Verstraete, Anderson, St. Louis, and Hudson (2018), Matos De Carvalho and Kingston (2018)). Nevertheless, in the absence of publicly available operational records on in-orbit servicing missions, these durations should be regarded as reasonable estimates rather than precise operational figures.

Finally, the refuelling anticipation window reported in Table 4.2 instantiates the preventive mechanism introduced in Section 3.1.3: the servicer may begin refuelling a client satellite up to $H = 50$ days before its fuel-failure threshold is reached. Within this window the satellite is still operational and continues to generate revenue; the revenue stream is interrupted only once the threshold itself is crossed. Preventive refuelling inside the window therefore avoids the upcoming service outage (and the associated forgone revenue) entirely, whereas the servicer remains free to intervene after the threshold, at the cost of the revenue lost in the interim.

4.1.4 Refuelling Depot Assumptions

The economic parameters adopted for the refuelling depot are summarised in Table 4.5. The depot price of \$150 M is treated as a capital expenditure that bundles both the construction (manufacturing) cost of the depot spacecraft and its launch and orbit-insertion cost; it is amortised over

the depot’s 15-year design life, each case study being charged only the share corresponding to its own horizon (Section 3.3). The station-keeping cost of \$12,000 per day represents the propellant and ground-operations expense incurred to hold the depot on its orbital slot over the mission horizon. As with the other economic quantities used in this study, both figures should be regarded as representative orders of magnitude rather than precise operational values. They are nonetheless consistent with the assumptions adopted in comparable on-orbit servicing studies like the one of Sarton du Jonchay et al. Sarton Du Jonchay et al. (2021, 2022).

Parameter	Value
Price [\$M]	150
Design life [years]	15
Station keeping [\$ /days]	12,000

Table 4.5: Economic cost of the refuelling depot

4.1.5 Servicer Assumptions

Table 4.6 summarizes the servicer configuration for both the low-thrust and high-thrust propulsion modes. These two configurations are applied consistently across all case studies to enable a direct comparison between propulsion architectures.

Parameter	Low-thrust servicer	High-thrust servicer
Propellant type	Xenon gas	Bi-propellant (MMH)
Propellant cost [\$/kg]	15,000	13,235
Dry mass [kg]	2,000	2,000
Fuel capacity [kg]	500	1,500
Build cost [\$M]	75	75
Build-and-launch cost [\$M]	107.5	120.5
Specific impulse [s]	{1500, 2000, 2500}	320
Power [W]	2,500	–
Thrust [N]	–	300
Starting True Anomaly [°]	0.0	0.0

Table 4.6: Servicer configuration for the low-thrust and high-thrust propulsion modes.

As mentioned in section 3.2.2, in low-thrust configuration, the reinforcement learning agent selects the propulsion operating point of every transfer it initiates. Indeed, at each decision epoch the agent chooses among three specific impulse values, uniformly spaced over the thruster’s operating range: $I_{sp} = 1,500$ s, corresponding to faster but more fuel-intensive manoeuvres; $I_{sp} = 2,000$ s, representing a balanced trade-off between transfer duration and propellant consumption; and $I_{sp} = 2,500$ s, corresponding to slower but more fuel-efficient manoeuvres. Following the action-space definition of Section 3.2.2, the low-thrust action space thus contains $3(|\mathcal{S}| + |\mathcal{D}|) + 1$ discrete actions. The number of selectable I_{sp} values is a tunable parameter of the framework and can be increased to provide finer control resolution if desired.

The high-thrust servicer uses Monomethyl Hydrazine (MMH) bi-propellant, while the low-thrust servicer uses xenon gas, with respective costs of 13,235 and 15,000 \$/kg. These figures correspond to the delivered cost of the propellant rather than its ground price: they combine the raw propellant cost, estimated at 235 \$/kg for MMH and 2,000 \$/kg for xenon, with the launch and orbit-delivery cost, assumed to be 13,000 \$/kg. This choice reflects the fact that every kilogram of propellant expended by the servicer must eventually be replaced in orbit, so its true economic

value is the cost of placing it there. These cost estimates are used to evaluate the fuel expenditure component r_t^{fuel} of the reward function defined in Section 3.2.3.

To make the propulsion trade-off concrete, Tables 4.7 and 4.8 report the average travel time and propellant cost of each transfer family between the three study-case orbits, for the high-thrust and low-thrust servicers respectively, evaluated for the servicer configurations defined above and averaged over all relative phasings between the servicer and its target. They quantify the defining contrast between the two architectures: the high-thrust servicer reaches any orbit within a few days, but spends several times more propellant than the low-thrust servicer, over an order of magnitude more for intra-orbit phasing, whereas the low-thrust Edelbaum spirals are far cheaper, yet require up to roughly three months for the inclination-changing transfers.

Table 4.7: Average travel time t , propellant mass m_p , and propellant cost of the high-thrust (HT) orbital manoeuvres between the three study-case orbits. Values are averaged over all relative phases. Servicer wet mass 3500 kg.

Transfer	Change	manoeuvre	t (d)	m_p (kg)	Cost (M\$)
Phasing (intra-orbit)	–	Phasing HT	4.0	103	1.36
o1 $\leftrightarrow$ o3	Δa (≈ 278 km)	Hohmann	4.6	56	0.74
o2 $\leftrightarrow$ o3	Δi (3°)	Plane change	4.0	384	5.08
o1 $\leftrightarrow$ o2	$\Delta a + \Delta i$	Hohmann + plane change	4.6	318	4.21

Table 4.8: Average travel time t and propellant cost of the low-thrust (LT) orbital manoeuvres between the three study-case orbits. Values are averaged over all relative phases. LT figures span the three available specific-impulse (throttle) settings, $I_{sp} = 1500\text{--}2500$ s, where lower I_{sp} gives faster manoeuvres and higher I_{sp} gives more propellant-efficient manoeuvres. LT inter-orbit transfers use continuous Edelbaum spirals. Servicer wet mass 2500 kg.

Transfer	Change	manoeuvre	t (d)	Cost (M\$)
Phasing (intra-orbit)	–	Phasing LT	11–14	0.05–0.10
o1 $\leftrightarrow$ o3	Δa (≈ 278 km)	Edelbaum spiral	8–12	0.08–0.13
o2 $\leftrightarrow$ o3	Δi (3°)	Edelbaum spiral	52–88	0.65–1.08
o1 $\leftrightarrow$ o2	$\Delta a + \Delta i$	Edelbaum spiral	54–88	0.66–1.09

4.2 Study case Results

All reinforcement learning models presented in this section were trained on an Apple M4 Pro processor with 24 GB of RAM. The orbital environment was implemented as a Semi-Markov decision process using the Gymnasium library, and the Maskable PPO agent was trained using the Stable-Baselines3 framework. The agent’s hyperparameters were tuned heuristically, through a manual trial-and-error search carried out separately for the high-thrust and low-thrust configurations. The two propulsion modes ultimately share the same hyperparameter set, with the sole exception of the entropy coefficient, which is set slightly higher for the low-thrust agent to promote exploration over its larger action space. The complete list of hyperparameters is reported in Appendix A.

4.2.1 Illustrative Example

Before analysing how the various case-study parameters affect performance, we first consider a deliberately simple scenario that illustrates how the trained agent makes its decisions and what the resulting servicing policy looks like. The scenario reuses the three MEO orbits introduced in Section 4.1.1, with seven client satellites distributed across them and a single refuelling depot placed

on orbit 1^M at a true anomaly of 90.0° . The servicer departs from orbit 1^M at 0.0° true anomaly. The satellites' initial operating times and the parameters of their Weibull laws are deliberately calibrated so that every satellite develops a service need early in the episode, making the agent's decision logic (and its effect on the environment) easy to observe. Figures 4.2, 4.3, 4.4 and 4.5 show the learned policies found by the HT and LT agents for this small case study.

Figures 4.2 and 4.3 trace the sequence of decisions taken by the HT agent over time. Starting from its initial position, the servicer performs a series of orbital manoeuvres to reach the satellites that require servicing soonest. The HT servicer fully exploits its defining advantage: very short transfer times, with no manoeuvre exceeding about eight days in this example, at the cost of a much higher propellant consumption than its LT counterpart. This physical advantage allows it to refuel satellite s_5 very early, within the preventive refuelling window, that is, before s_5 develops an actual refuelling need (shown in yellow) and therefore before it stops generating revenue. The servicer then continues to service the remaining satellites as their needs arise, returning three times to the depot (green stars in the figure) to maintain a propellant level that allows it to keep operating. Across the campaign, the agent draws on the full repertoire of high-thrust manoeuvres available to it, i.e., phasing, Hohmann, Hohmann with inclination change, and plane change, as reflected by the different transit colours in Figure 4.2. Once a satellite has been serviced, its timeline turns to a shaded grey band ("Serviced"), indicating that its need has been met and that it will keep generating revenue until the end of the horizon. As shown in Figure 4.3, once the last satellite has been serviced (the deorbiting of s_4) the servicer no longer has any service to perform and issues "wait" actions in a loop until the end of the horizon. This prevents any further propellant consumption, an optimal choice that maximises the economic value of its action sequence.

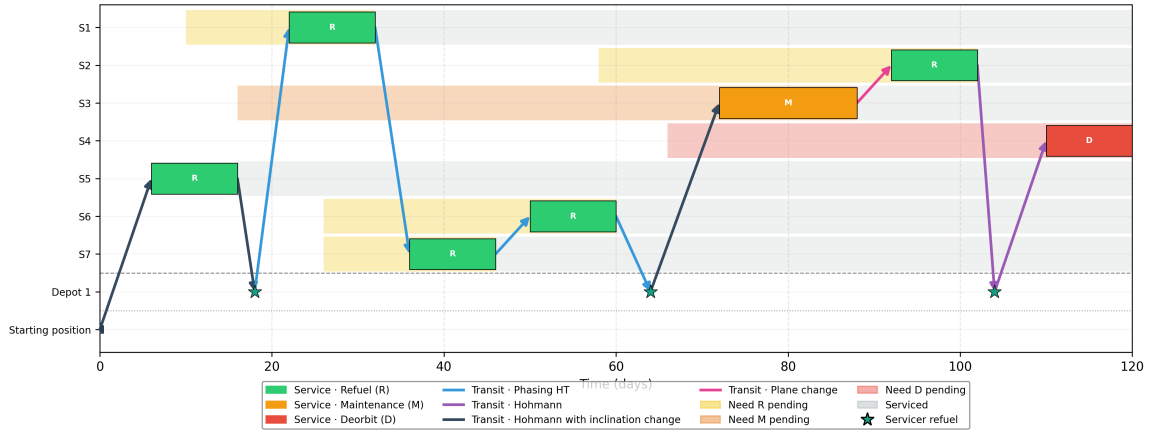


Figure 4.2: Operation timeline of the high-thrust (HT) servicer over days 0–120 of the showcase scenario.

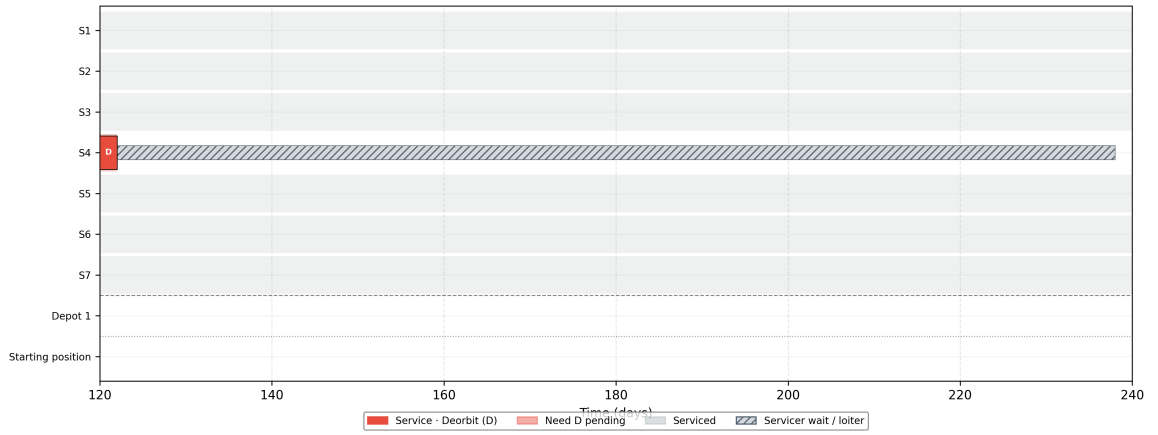


Figure 4.3: Operation timeline of the high-thrust (HT) servicer over days 120–240 of the showcase scenario.

In a similar manner, Figures 4.4 and 4.5 show the action sequence taken by the LT servicer over the course of the simulation. The transfer durations are, on average, markedly longer than those of the HT servicer, which is consistent with the low-thrust philosophy: fuel-efficient but time-consuming. As a direct consequence, the LT agent is unable to refuel the satellites within their preventive windows as the HT agent did for s_5 . Instead, several satellites cross their fuel-failure threshold before the servicer can reach them and enter the “Need R pending” state (shown in yellow), losing revenue until the corresponding service is completed. This effect is most striking

for s_5 : refuelled preventively around day 10 by the HT servicer, it is left pending from roughly day 35 until about day 178 in the LT case, a long revenue outage. Satellite s_7 likewise accumulates an extended pending-refuel interval before being serviced. The slower pace is equally apparent in the overall completion time: the LT servicer finishes its last service, the refuelling of s_5 , around day 190 and remains occupied almost until the end of the mission, whereas the HT servicer had completed the entire campaign by about day 120 and then loitered for the remainder of the horizon. Finally, it is worth noting that, in this small scenario, the LT servicer never needs to visit the depot: its high propellant efficiency allows it to complete the whole campaign on its initial fuel load, in contrast to the three refuelling stops required by the HT servicer.

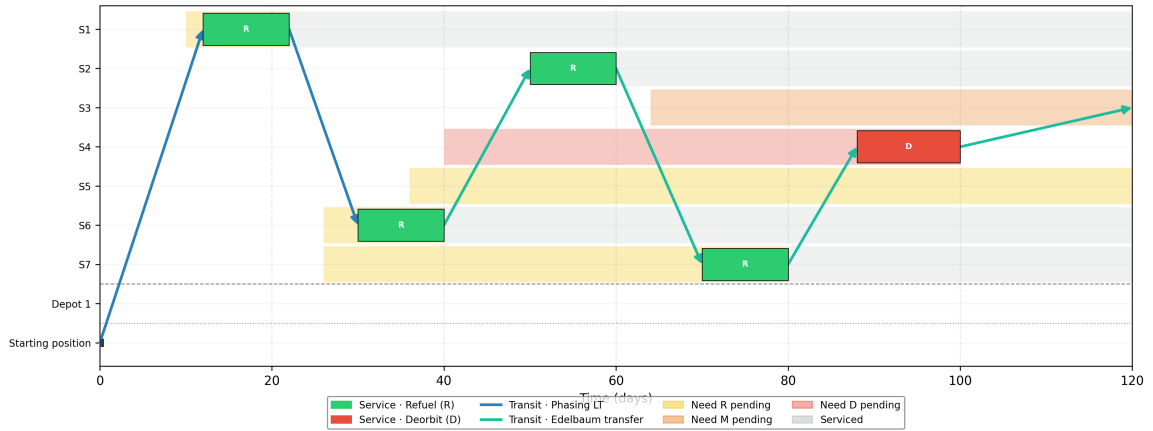


Figure 4.4: Operation timeline of the low-thrust (LT) servicer over days 0–120 of the showcase scenario.

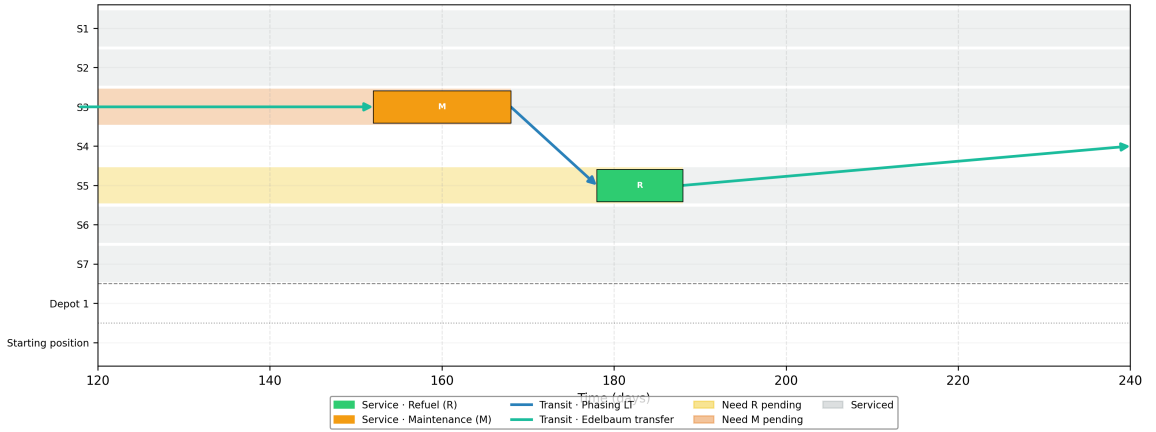


Figure 4.5: Operation timeline of the low-thrust (LT) servicer over days 120–240 of the showcase scenario.

4.2.2 Short-Horizon Results

We now turn to the case study combining a short mission horizon (one year) with a small service demand (30 client satellites). The HT agent was trained for 30×10^6 steps and the LT agent for 35×10^6 steps, requiring respectively 143 and 209 minutes of wall-clock time; the longer training time of the LT agent is consistent with its larger action space, which offers three selectable specific-impulse settings per candidate target.

All monetary quantities reported in this section follow the economic model of Section 3.3: every cash flow is discounted at $\delta = 8\%$ per year (Eq. 23), capital expenditures are booked at $t = 0$ with the depot price prorated over its 15-year design life, i.e., \$10.0M share for the one-year horizon considered here, and replaced satellites are charged the prorated, discounted cost of Eq. 26.

Figure 4.6 provides a decomposed economic comparison between the HT and LT servicers under these conventions. The three leftmost groups quantify the costs incurred by the servicing system. The “initial investment” combines the build-and-launch cost of the servicer with the prorated depot share of \$10.0M. The servicer cost itself comprises a \$75M platform (manufacturing) cost plus a launch cost of \$13,000 per kilogram of wet mass, the same specific launch cost underlying the delivered-propellant prices of Section 4.1.5. With a 2,000 kg dry mass and full propellant loads of

500 kg (LT) and 1,500 kg (HT), this yields \$107.5M for the LT servicer and \$120.5M for the HT servicer, hence total initial investments of \$117.5M and \$130.5M, respectively. The "operational cost" combines the daily station-keeping cost of the depot, discounted day by day, with the propellant cost expended by the servicer, the split between these two contributions being shown in the inset. The "asset loss" accounts for the prorated cost (Eq. 26) of the satellites lost to deorbiting failures. On the positive side, the "satellite fleet revenue" reports the discounted revenue generated by the operational satellites over the whole mission horizon, and the rightmost group reports the resulting net present value (NPV), defined as the fleet revenue minus the three cost groups. All stochastic quantities are averaged over 1000 independent evaluation episodes.

The defining philosophy of each architecture: time-efficient but fuel-consuming for the HT servicer, fuel-efficient but time-consuming for the LT servicer, is clearly reflected in Figure 4.6. The HT servicer (orange bars) satisfies the fleet's service needs faster than the LT servicer, which yields a slightly higher fleet revenue (\$1217.4M against \$1183.8M, i.e. a \$33.6M advantage). This speed, however, is paid for twice. First, in propellant: the HT servicer burns \$19.8M of propellant against only \$2.3M for the LT servicer, more than eight times as much, while the depot station-keeping cost is identical for both architectures (\$4.2M each), so the entire operational-cost gap stems from propellant expenditure alone. Second, in capital expenditure: the larger propellant load of the HT servicer (1,500 kg against 500 kg) makes it \$13.0M more expensive to launch. Asset losses are equal (\$5.9M for both HT and LT), reflecting the few deorbiting failures that occur over such a short horizon. In total, the \$33.6M revenue gain of the HT servicer is almost exactly offset by its \$17.5M of additional propellant cost and its \$13.0M of additional initial investment, leaving the two architectures with nearly identical NPVs of \$1053.8M (LT) and \$1057.0M (HT), a \$3.2M difference, about 0.2%, well within the episode-to-episode variability. For this particular case study, the LT and HT servicers are therefore economically equivalent options.

This result was expected. The on-orbit service demand is not high enough to justify the fast, but costly, transfers of the HT servicer. As the demand increases, we expect the LT servicer to become progressively unable to keep pace, taking longer to service satellites on time; this would raise the fraction of satellites left in a failed state over the horizon and, in turn, erode the revenue of the LT configuration.

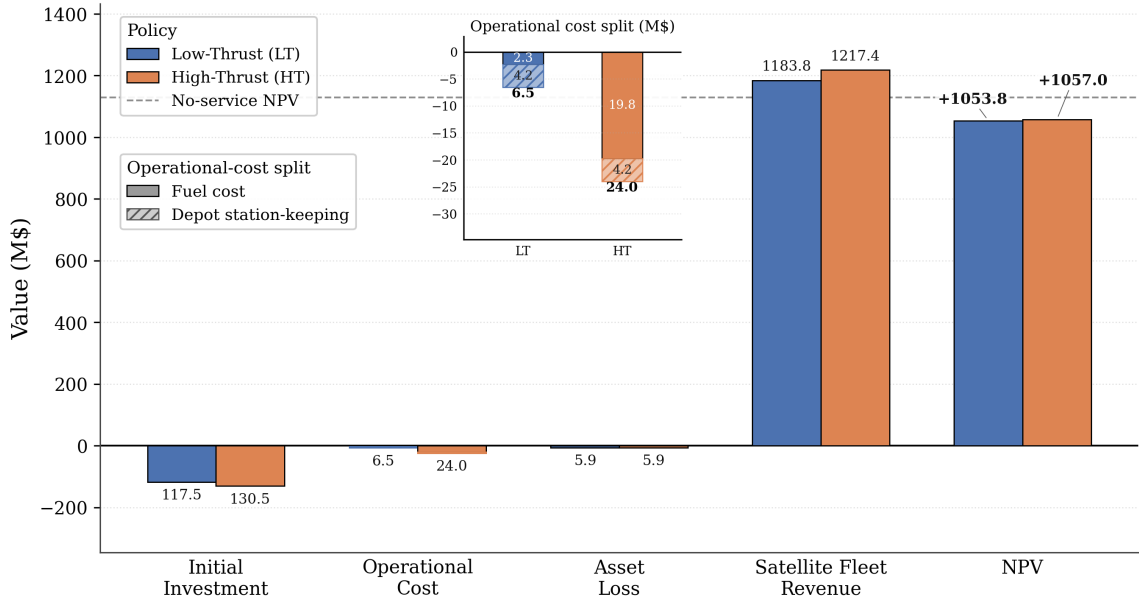


Figure 4.6: Economic comparison between the high-thrust (HT) and low-thrust (LT) servicers for the short-horizon, small-demand case study (one-year horizon, 30 satellites). The inset details the split of the operational cost into propellant cost and depot station-keeping, and the dashed line marks the NPV of the no-service baseline of Figure 4.7. All cash flows are discounted at $r = 8\%$ per year and averaged over 1000 evaluation episodes.

Figure 4.7 compares the NPV of the HT and LT servicers against a baseline reflecting current industry practice in the absence of on-orbit servicing: a satellite is replaced as soon as it develops a service need, irrespective of the type of need. To capture the procurement and launch lead time of a replacement spacecraft, the affected orbital slot remains dark, generating no revenue, for 90 days before the fresh satellite enters service. The replacement cost is prorated according to Eq. 26 and discounted like every other cash flow. The same replacement rule applies under the HT and LT servicing policies, with the sole difference that a satellite is then replaced only when it reaches a deorbiting need. The no-service baseline, finally, deploys neither servicer nor depot and therefore bears no infrastructure cost.

Figure 4.7 shows that the one-year horizon and the small fleet (30 satellites) do not allow either servicing model to amortise its infrastructure. Denoting by $\Delta\text{NPV} = \text{NPV}_{\text{service}} - \text{NPV}_{\text{no-service}}$

the value created relative to the no-service baseline, both policies fall short of the \$1128.9M no-service level, with $\Delta\text{NPV} = -\$75.0\text{M}$ for the LT servicer and $-\$71.8\text{M}$ for the HT servicer. In this low-demand, short-horizon regime, deploying a dedicated servicing infrastructure is thus not yet economically justified relative to simply replacing satellites upon failure. The shortfall nonetheless remains limited to about 6–7% of the no-service NPV, and stems essentially from an infrastructure that a single year of operation, under such a low service demand, gives the servicer too few opportunities to amortise. As noted above, this balance is expected to shift in favour of the servicing models as the service demand and the mission horizon increase, which we investigate next.

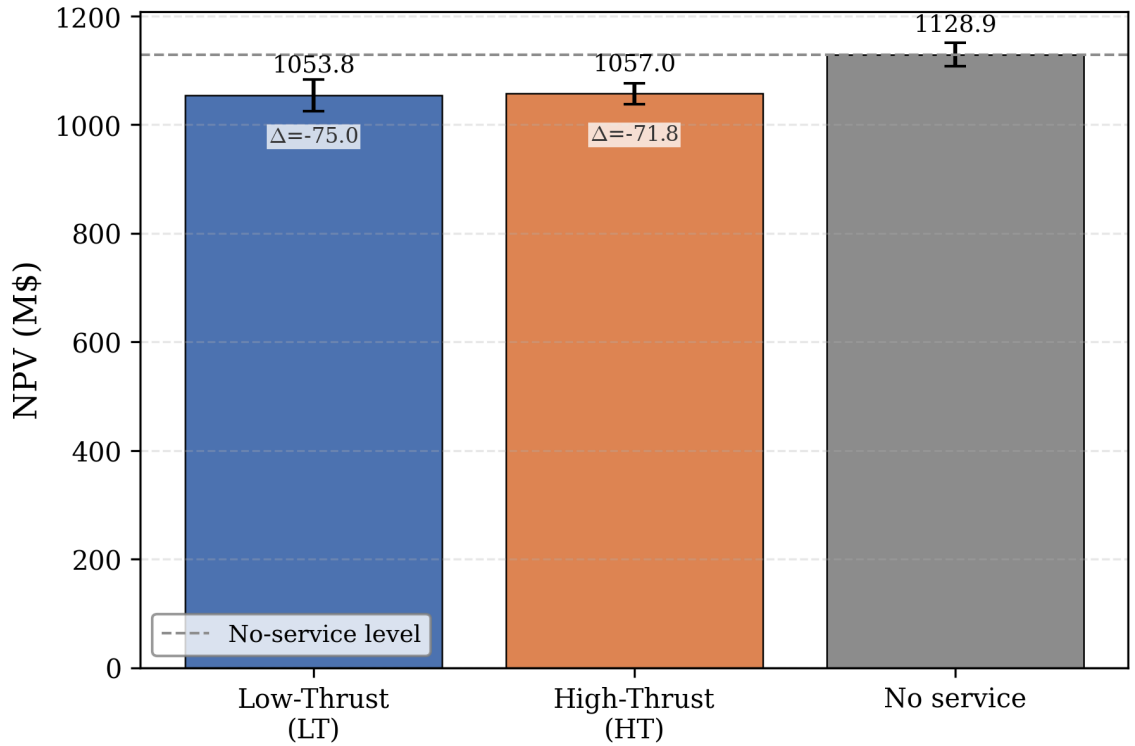


Figure 4.7: NPV comparison between the servicer models (HT and LT) and the no-service baseline, in which any satellite that triggers a service need is immediately replaced. Error bars indicate the variability across the 1000 evaluation episodes, and Δ denotes the NPV gap relative to the no-service level.

4.2.3 Long-Horizon Results

We now extend the mission horizon to five years while keeping the on-orbit service demand unchanged, in order to assess whether the HT and LT servicing infrastructures become economically viable over a longer operating period. As shown in Section 4.2.2, a one-year horizon was too short for either the HT or the LT servicer to outperform the pure replace-on-failure policy that currently prevails in industry.

Figure 4.8 reports the decomposed economic results obtained by the HT and LT models for a five-year horizon and a fleet of 30 satellites. For this case study, the HT agent was trained for 35 million steps, corresponding to a wall-clock training time of 174 minutes, whereas the LT agent was trained for 50 million steps, corresponding to a wall-clock training time of 270 minutes.

On the cost side, the initial investment rises from \$117.5M and \$130.5M in the short-horizon case to \$157.5M (LT) and \$170.5M (HT): the servicer build-and-launch costs are unchanged, but the five-year window is now charged \$50.0M of the \$150M depot price, one third of its 15-year design life, instead of the \$10.0M share of the one-year study. Both servicers also expend far more propellant than over the short horizon, \$83.6M for the HT servicer and \$13.8M for the LT servicer, since a five-year horizon generates proportionally more service demand and therefore more transfers to perform. The HT servicer still burns about six times as much propellant as its LT counterpart, and since the discounted depot station-keeping cost is identical for the two architectures (\$18.2M each), the entire operational-cost gap (\$101.7M against \$32.0M) again stems from propellant expenditure alone.

On the revenue side, the picture is consistent with the short-horizon case and with the design philosophy of each architecture: the fast transfers of the HT servicer keep satellites operational (and revenue-generating) for longer, yielding a fleet revenue of \$5258.3M against \$5217.3M for the LT servicer, a \$41.0M advantage. The asset losses, in contrast, are no longer negligible: they reach \$117.6M for the LT servicer and \$116.5M for the HT servicer, reflecting the deorbiting failures that accumulate as the fleet ages over five years, but they remain essentially identical for the two architectures and thus do not contribute to the gap between them. The decisive factor is, once again, the cost of speed: the \$41.0M revenue advantage of the HT servicer is more than offset by its \$69.8M

of additional propellant expenditure and its \$13.0M of additional initial investment, leaving the LT servicer with a modestly higher NPV of \$4910.2M against \$4869.6M, a \$40.6M gap, i.e. less than 1%. Over a long horizon and under a small service demand, the propellant economics therefore tip the balance in favour of the fuel-efficient LT architecture, although both options remain remarkably close.

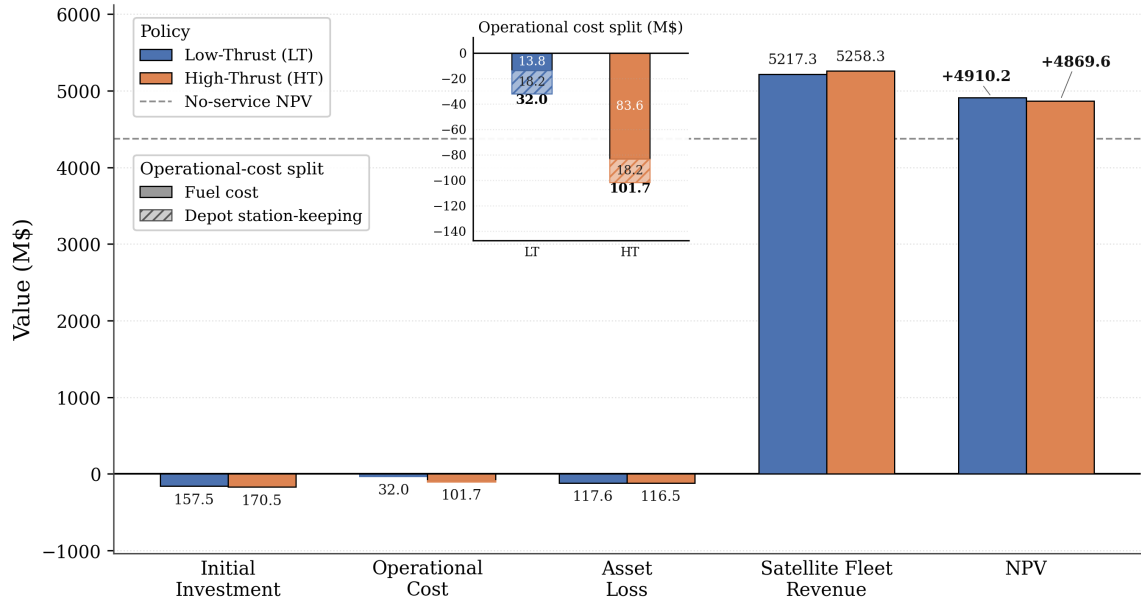


Figure 4.8: Economic comparison between the high-thrust (HT) and low-thrust (LT) servicers for the long-horizon, small-demand case study (five-year horizon, 30 satellites). The inset details the split of the operational cost into propellant cost and depot station-keeping, and the dashed line marks the NPV of the no-service baseline. All cash flows are discounted at $r = 8\%$ per year and averaged over 1000 evaluation episodes.

Figure 4.9 shows the fraction of the fleet that remains operational over time, i.e. the satellites that generate revenue and have no pending or in-progress service need, for the two servicing policies and for the no-service baseline. The horizon-averaged uptime reaches 98.7% for the HT servicer, 97.6% for the LT servicer, and 96.0% for the no-service baseline. The ranking mirrors the fleet-revenue results: the HT servicer sustains the highest availability, which is precisely the operational advantage its propellant expenditure buys. The differences are most pronounced during the first year of the mission, when the initial wave of refuelling deadlines falls due: the slower LT servicer

temporarily falls behind and its availability dips to about 92–93%, while the no-service baseline drops as low as 86%, each unattended need leaving an orbital slot dark for the 90-day replacement lead time. Both servicing policies keep the fleet above the no-service level over essentially the whole horizon. It is worth emphasising, however, that the roughly one additional point of uptime delivered by the HT servicer translates into the \$41.0M revenue advantage discussed above is not enough, under this small demand, to pay for its propellant cost.

The propellant bill of the HT servicer is dominated by depot access: whenever a refuelling trip requires an inclination change, the manoeuvre costs on average 300–380 kg of propellant, roughly 20–25% of a full tank, or \$4–5M per visit (Table 4.7). This cost could be substantially reduced by increasing the number of refuelling depots: placing one depot per orbit, for example, would let the HT servicer reach a depot through a simple phasing manoeuvre or a Hohmann transfer, both far cheaper than the inclination changes that currently dominate its depot-access cost. This option is investigated in Section 4.2.5.

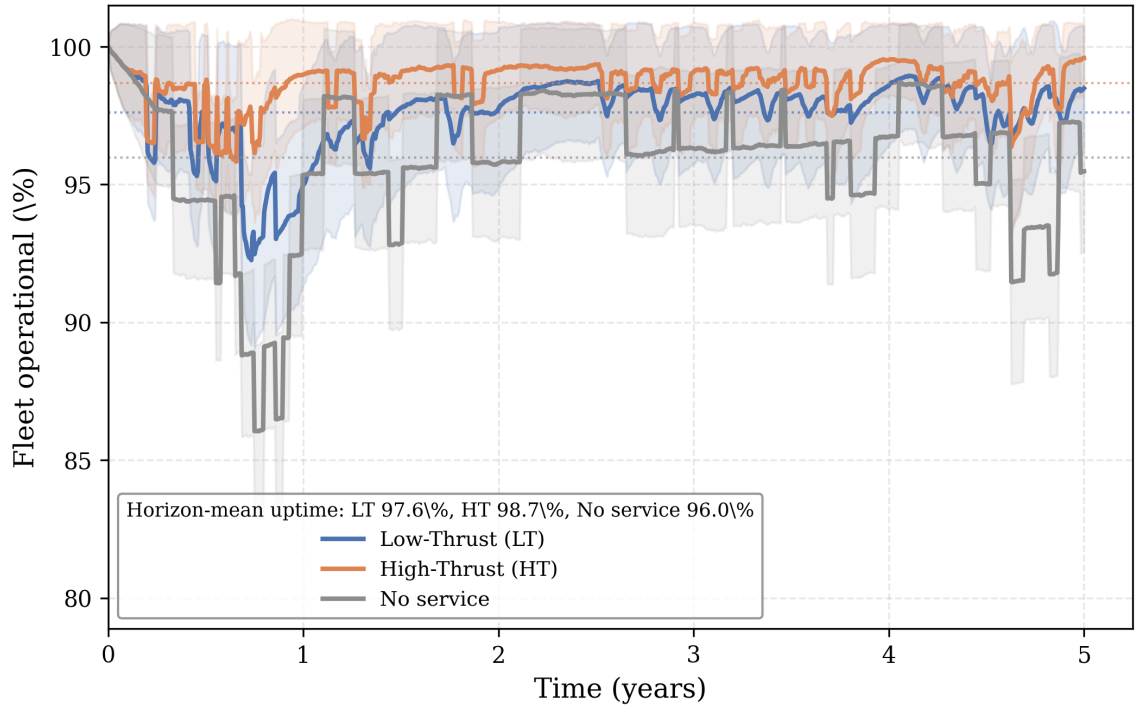


Figure 4.9: Fraction of the satellite fleet that is operational, i.e. generating revenue, with no pending or in-progress service need, over the five-year horizon, for the high-thrust (HT) and low-thrust (LT) servicers and the no-service baseline in the small-demand case study. Solid lines show the mean across the 1000 evaluation episodes and the shaded bands the corresponding spread; the horizontal dotted lines mark the horizon-averaged uptimes (HT 98.7%, LT 97.6%, no service 96.0%).

Figure 4.10 plots the difference between the cumulative discounted NPV of each servicing policy (HT and LT) and that of the no-service baseline, the time-resolved counterpart of the NPV gap ΔNPV introduced for the short horizon in Figure 4.7. Each curve starts at the negative of the corresponding initial investment (\$157.5M for LT, \$170.5M for HT, booked at $t = 0$) and rises as the avoided service outages accumulate. Both servicing policies overtake the no-service baseline well within the first year, after roughly seven months of operation for the LT servicer and eight months for the HT servicer, the slightly earlier break-even of the LT policy reflecting its lighter, cheaper-to-launch configuration. Value then accrues steeply over the first two years, when the initial wave of service needs falls due, and more gradually thereafter. By the end of the horizon, the servicing

policies create \$537M (LT) and \$496M (HT) of value relative to the no-service NPV of approximately \$4373M, a gains of roughly 12% and 11%, respectively. This stands in sharp contrast to the one-year study, and confirms that the profitability of a dedicated servicing infrastructure is primarily a matter of operating timescale. Finally, the HT curve saturates, and even slightly declines, over the final year of the mission. This is an end-of-horizon effect: as the remaining operating window shrinks, the discounted revenue recoverable from servicing an additional satellite no longer justifies the propellant cost of reaching it, so late interventions cease to create value relative to simply letting the baseline replace the satellite.

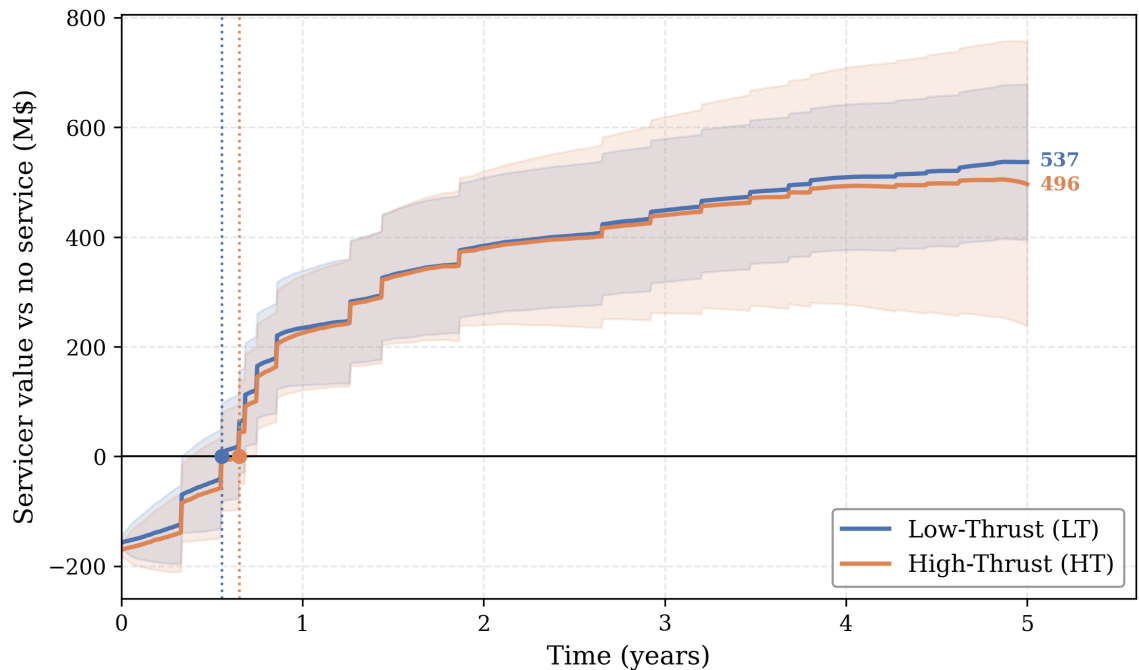


Figure 4.10: Servicer value relative to the no-service baseline as a function of mission time, for the long-horizon, small-demand case study (five-year horizon, 30 satellites). Each curve is the cumulative discounted NPV of a servicing policy (HT or LT) minus that of the no-service baseline; the dotted vertical lines and markers indicate the break-even epoch at which each policy overtakes the no-service strategy. Solid lines show the mean over the 1000 evaluation episodes and the shaded bands the corresponding spread.

4.2.4 Impact of the Service Demand

For this section, we adopt a five-year horizon throughout. As the short-horizon analysis of Section 4.2.2 made clear, a one-year horizon is too short for a dedicated servicing infrastructure to amortise its initial investment, and a multi-year timescale is far more representative of the operational economics of on-orbit servicing in MEO, where satellites remain in service for ten to fifteen years. Within this setting, we now investigate how the volume of service demand affects the relative performance of the HT and LT servicers. To this end, we vary the size of the client fleet across 30, 50, 70, and 90 satellites, holding all other parameters fixed.

Figure 4.11 reports the adjusted NPV of each servicing policy relative to the no-service baseline. This is the same NPV gap introduced in Section 4.2.2, namely the difference between the servicer NPV and the no-service NPV, $\Delta\text{NPV} = \text{NPV}_{\text{service}} - \text{NPV}_{\text{no-service}}$, evaluated under the discounted-cash-flow conventions of that section. A curve lying above zero therefore indicates that the corresponding servicing policy creates more value than the prevailing replace-on-failure strategy. In this study, both servicers operate with a single refuelling depot, so the comparison isolates the effect of fleet size alone.

Several observations emerge from Figure 4.11. First, the HT servicer overtakes the LT servicer at a fleet size of only about 34 satellites, and pulls away steadily beyond that point. This confirms the expectation formulated in Section 4.2.2: as service demand grows, the slower LT servicer becomes increasingly unable to reach satellites before they fail, so the time-efficient HT architecture pays off, and it does so at a demand level barely above the 30-satellite case studied so far. The HT value creation keeps rising with demand, from about \$500M at 30 satellites to a peak of approximately \$1360M at 70 satellites.

The trajectory of the LT servicer is markedly different. Its value creation peaks at about \$1040M for a 50-satellite fleet, erodes to roughly \$740M at 70 satellites, and collapses below the no-service baseline at 90 satellites (ΔNPV of approximately $-\$100\text{M}$). A single low-thrust servicer simply saturates: as unattended service needs accumulate, satellites remain non-operational far longer than the 90-day replacement lead time of the baseline, so that maintaining the servicing infrastructure ends up destroying value relative to simply replacing satellites upon failure. In the high-demand

regime, the fuel efficiency that makes the LT architecture attractive at low demand is no longer worth its slowness.

The HT servicer, by contrast, remains strongly profitable across the entire range considered, although its value creation also declines beyond 70 satellites, to approximately \$1060M at 90 satellites. At such demand levels, a single servicer drawing on a single depot begins to saturate as well: refuelling trips become more frequent, and with the sole depot located on a single orbital plane, reaching it from the other orbits requires costly inclination-change manoeuvres (Table 4.7). It is also worth noting that the episode-to-episode spread widens considerably with fleet size, at 90 satellites the LT band straddles zero, reflecting the growing number of stochastic failure events per episode. Overall, this sweep shows that transfer speed, rather than fuel efficiency, becomes the decisive attribute as demand grows, and it suggests that the depot infrastructure, rather than the propulsion mode itself, may be what ultimately limits the HT servicer at very high demand. This hypothesis is tested in the next section.

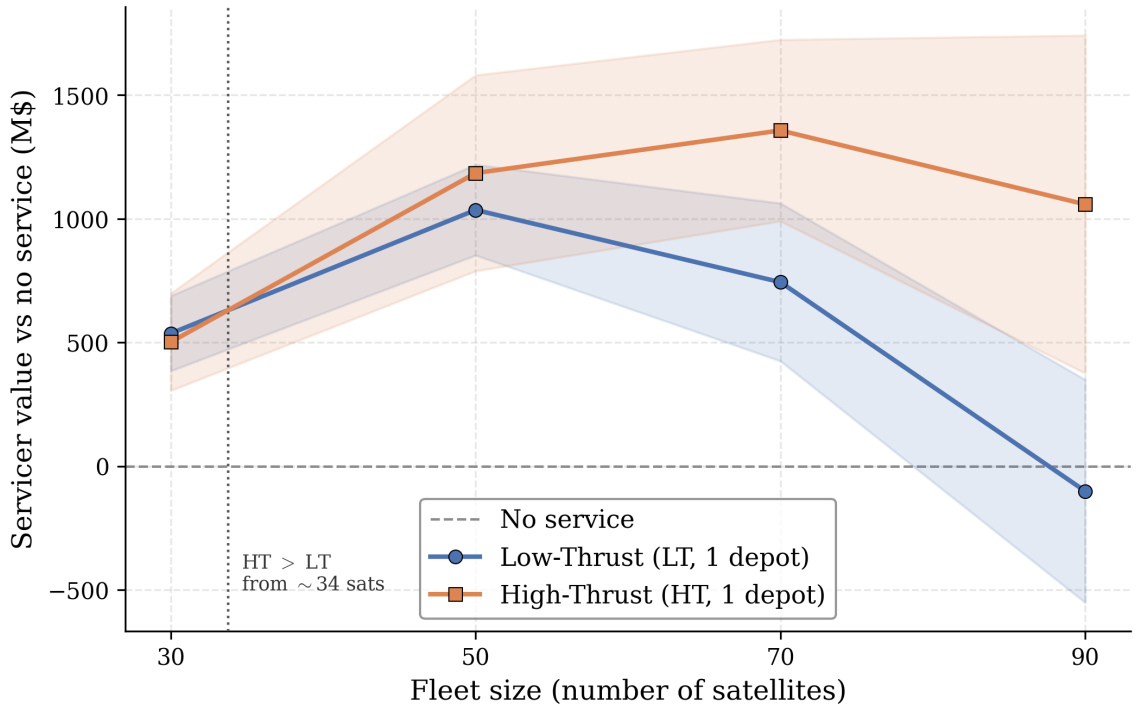


Figure 4.11: Adjusted NPV of the HT and LT servicing policies relative to the no-service baseline, $\Delta\text{NPV} = \text{NPV}_{\text{service}} - \text{NPV}_{\text{no-service}}$, as a function of fleet size for the long-horizon (five-year) case studies in which both servicers operate with a single refuelling depot. A value above zero indicates that the servicing policy creates more value than replacing satellites on failure. Markers show the mean over the 500 evaluation episodes and the shaded bands the corresponding spread; the vertical dotted line marks the fleet size (around 34 satellites) at which the HT servicer overtakes the LT servicer.

Nevertheless, Figure 4.11 neatly captures the strengths and weaknesses of each propulsion mode: time-efficient but fuel-consuming for the HT servicer, and fuel-efficient but time-consuming for the LT servicer. For small fleets, approximately 30 satellites or fewer, where service demand remains moderate, the LT servicer performs particularly well due to its low propellant consumption and limited need for refuelling. As demand increases, with fleets of 50, 70, and 90 satellites, the LT servicer increasingly suffers from its main drawback, i.e., long transfer times, and progressively fails to reach satellites before they fail. In contrast, the HT servicer comes into its own by exploiting its very short transfer times to keep pace with the higher service demand.

4.2.5 Impact of the Number and Position of Refuelling Depots

We now examine the effect of the number, and subsequently the position, of the refuelling depots on the economic performance of the LT and HT servicers. This finally lets us test the hypothesis raised in Section 4.2.3 and sharpened by the demand sweep of Section 4.2.4: that a richer depot infrastructure could relieve the propellant burden of depot access, in particular for the largest fleets (70 and 90 satellites).

Figure 4.12 plots the NPV gap between the servicing and no-service policies, ΔNPV , for three depot configurations. In the single-depot case, the depot is placed on orbit 1^M at a true anomaly of 290.6° ; in the two-depot case, the depots are placed on orbits 1^M and 2^M at initial true anomalies of 290.6° and 180° ; in the three-depot case, one depot is placed on each orbit, at initial true anomalies of 290.6° , 180° , and 0° , respectively. Each additional depot carries its own acquisition, launch, and station-keeping cost, charged according to the conventions of Section 3.3, so roughly \$68M per depot over the five-year window (\$50.0M of prorated capital expenditure and \$18.2M of discounted station-keeping). The single-depot curves reproduce those of Figure 4.11.

For the LT servicer (left panel), adding refuelling depots does not improve economic performance, on the contrary. At every fleet size, the two and three depot configurations yield a lower ΔNPV than the single-depot one, and at 90 satellites all three configurations fall below the no-service baseline. The only exception is that, for the 90-satellite fleet, the two-depot configuration (about $-\$20\text{M}$) edges out both the single-depot (about $-\$110\text{M}$) and three-depot (about $-\$270\text{M}$) ones; the margin, however, is small compared to the episode-to-episode spread and does not alter the overall picture. This behaviour is a direct consequence of the LT servicer's very low propellant consumption: it visits a depot only rarely, so the marginal servicing capacity unlocked by additional depots cannot offset the cost of building, launching, and maintaining them. The extra infrastructure is therefor essentially dead weight.

The HT servicer (right panel) follows a similar pattern, contrary to expectation. Adding depots does not raise its ΔNPV either: the best value is reached at 70 satellites, where the one- and two-depot configurations are essentially tied at about \$1360M, and at 90 satellites the two-depot configuration exceeds the single-depot one only marginally (about \$1070M against \$1060M), while

three depots are consistently the worst option. The explanation is geometric. In these case studies, the three MEO orbits (Table 4.1) are nearly coplanar: only orbit 2^M has a non-zero inclination, and the maximum inclination separation is 3° . Since inclination changes are by far the most expensive manoeuvres (Table 4.7), while phasing and Hohmann legs remain cheap, even the single-depot agent learns to keep its depot-access cost low, batching several services between refuelling trips, so the propellant saved by additional depots is too small to repay their infrastructure cost. Two conclusions follow. First, the depot-bottleneck hypothesis does not hold in this near-coplanar geometry: the residual decline of the HT value beyond 70 satellites is better explained by the throughput saturation of a single servicer than by depot access. Second, the value of a depot network should grow with the geometric separation of the orbits it serves. To test this, the remainder of this section considers a constellation whose orbits are markedly more separated.

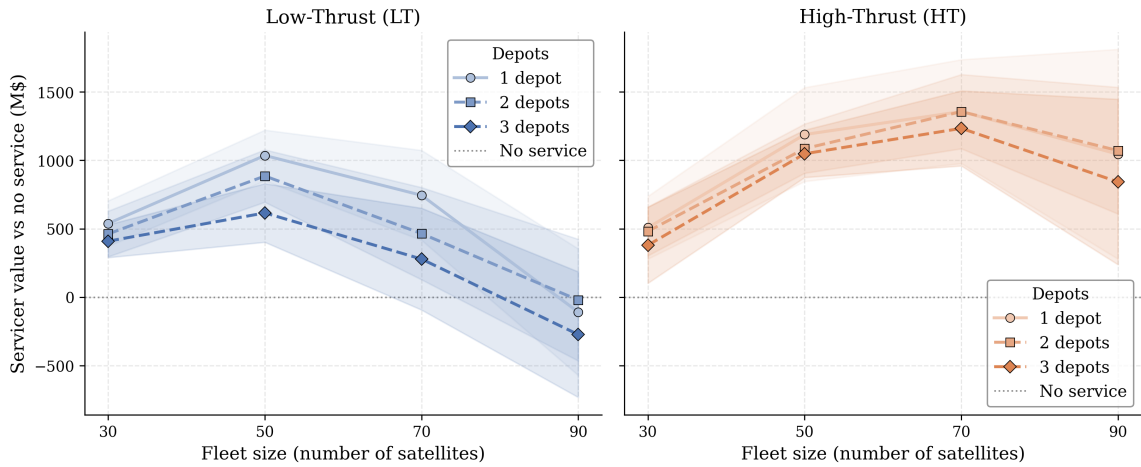


Figure 4.12: Effect of the number of refuelling depots on the adjusted NPV (ΔNPV , servicer value relative to the no-service baseline) as a function of fleet size, shown separately for the low-thrust servicer (left) and the high-thrust servicer (right). In each panel the one, two, and three depot configurations are compared, with the dotted line marking the no-service baseline. Markers show the mean over the 500 evaluation episodes and the shaded bands the corresponding spread. In this near-coplanar geometry, additional depots degrade the economics of both servicers.

The new constellation geometry is reported in Table 4.9. Only orbit 3^M is modified: its inclination is raised to 6° and its orbital period to 18100 s, i.e., a semi-major axis of 14899.7 km, so

that the three planes are now mutually separated by 3° of inclination, with a worst-case separation of 6° between orbits 1^M and 3^M . In addition, the propellant capacity of the HT servicer is reduced to 800 kg, deliberately forcing more frequent depot visits. These orbits support three case studies with one, two, and three refuelling depots, placed on one, two, or all three of the orbits at the same true anomalies as before. The mission horizon is set to two years and the client fleet comprises 24 satellites evenly distributed across the three orbits (eight per orbit). Only the HT servicer is studied here: being by far the more propellant-hungry architecture, it is the one most sensitive to the depot infrastructure.

Orbit id	Period (sec)	Semi major axis (km)	Inclination (deg)	RAAN (deg)
1^M	17100	14345.8	0.0	0.0
2^M	17600	14624.1	3.0	0.0
3^M	18100	14899.7	6.0	0.0

Table 4.9: Orbital parameters used for the depot-placement case studies

Figure 4.13 reports the Δ NPV achieved by the HT servicer with one, two, and three refuelling depots in this modified geometry. This time, the number of depots has a significant impact. The single-depot configuration, i.e., the depot sitting on orbit 1^M , is the weakest, at \$73M. The value peaks at \$214M with two depots, on orbits 1^M and 2^M , and then recedes to \$132M with three. These results confirm that increasing the number of depots does raise the servicer value when the orbits are sufficiently separated for inter-orbit manoeuvres to be expensive. With a single depot on orbit 1^M , every refuelling trip from orbit 3^M requires a costly 6° inclination change. Adding a depot on orbit 2^M caps the worst-case plane change at 3° from anywhere in the constellation, which yields substantial propellant savings; this is where the peak occurs. A third depot, in contrast, no longer reduces the worst-case plane change, it only adds convenience, so the propellant it saves does not repay its own acquisition, launch, and station-keeping costs: the three-depot configuration remains well above the single-depot one, but below the two-depot optimum.

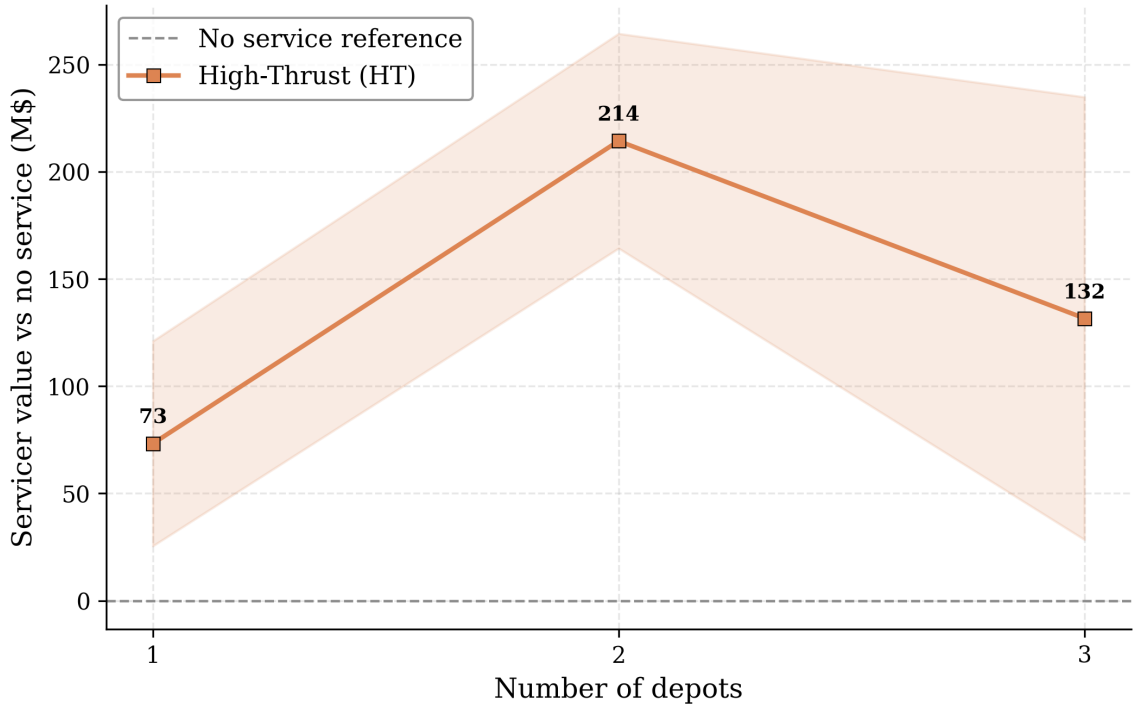


Figure 4.13: Effect of the number of refuelling depots on the adjusted NPV of the high-thrust servicer (Δ NPV, servicer value relative to the no-service baseline) for the modified constellation geometry of Table 4.9 (two-year horizon, 24 satellites, 800 kg propellant capacity). Markers show the mean over the 1000 evaluation episodes and the shaded bands the corresponding spread.

Figure 4.14 isolates the effect of depot placement: it reports the Δ NPV of the single-depot configuration when the depot's initial orbit is varied across the three planes. The peak occurs with the depot on orbit 2^M , at \$209M, while placements on orbits 1^M and 3^M yield comparable results of \$73M and \$89M, respectively. The explanation mirrors the previous one: orbit 2^M carries the intermediate inclination, so a depot placed there is reachable from anywhere in the constellation with at most a 3° plane change, whereas a depot on orbit 1^M or 3^M leaves a worst-case change of 6° . Strikingly, this single well-placed depot (\$209M) captures essentially the entire value of the best two-depot configuration (\$214M) while carrying half the depot infrastructure: in this scenario, placement quality almost fully substitutes for depot count.

These two figures carry the central message of this section. Expanding the depot network creates value only in scenarios where inter-orbit manoeuvres are expensive; when the constellation is nearly

coplanar, additional depots are dead weight for both propulsion architectures. And the number of depots is not the only design lever: the initial placement of the depots matters just as much for the economic viability of the servicing infrastructure, with the best single placement rivalling the best multi-depot configuration at a fraction of the cost.

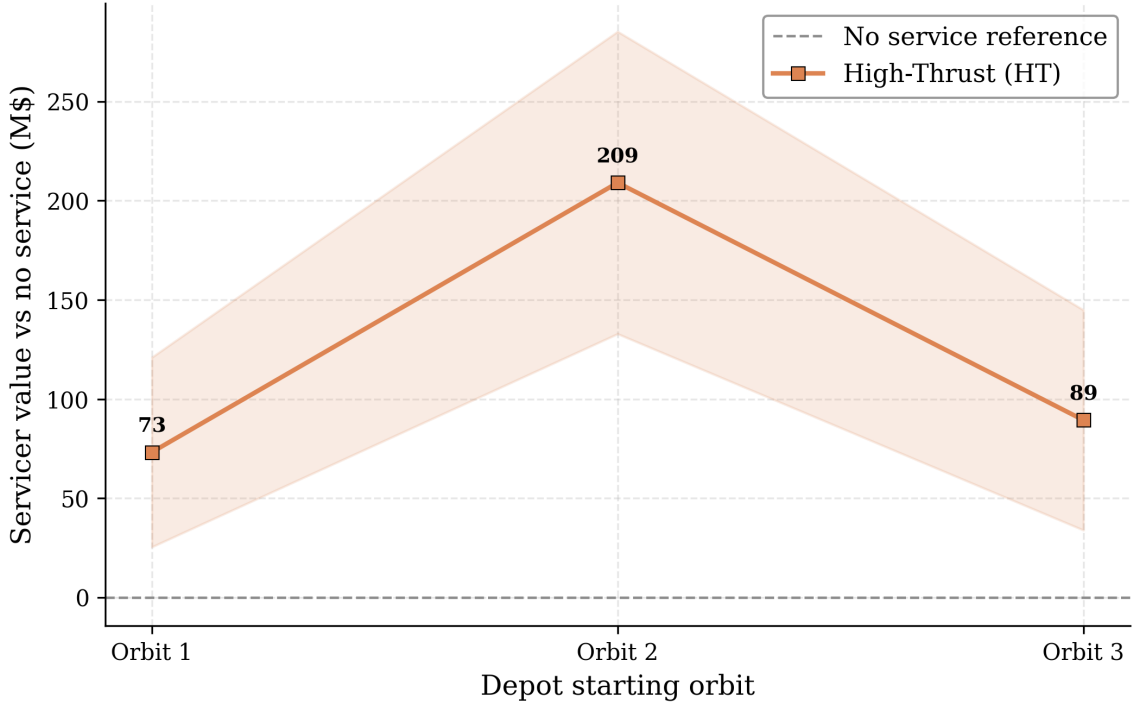


Figure 4.14: Effect of the initial orbital placement of a single refuelling depot on the adjusted NPV of the high-thrust servicer (ΔNPV , servicer value relative to the no-service baseline) for the modified constellation geometry of Table 4.9. Markers show the mean over the 1000 evaluation episodes and the shaded bands the corresponding spread.

4.2.6 Comparison of the RL and Greedy Policies

We conclude this section by comparing the trained reinforcement-learning policies against the greedy value-at-stake heuristic, following the protocol of Section 3.5: the two policies drive the same servicer through the same environment, receive the same observations, are constrained by the same feasibility mask, and are evaluated on the same stochastic seeds, so that any performance gap is attributable to the quality of their decisions alone. Figure 4.15 reports the adjusted NPV, ΔNPV ,

of both policies for the fleet sizes of Section 4.2.4 (single depot, five-year horizon), the low-thrust servicer on the left panel and the high-thrust servicer on the right; the RL curves are those of Figure 4.11.

For the LT servicer, the learned policy dominates the heuristic at every demand level, and the gap widens dramatically with fleet size. The two policies are on par for the 30-satellite fleet (about \$540M each): the demand is sparse enough for even a myopic dispatcher to keep up. Beyond that point, the heuristic collapses. It barely breaks even at 50 satellites (about \$50M, against \$1040M for the RL policy), destroys value from 70 satellites onward (about $-\$900\text{M}$ against $+\$740\text{M}$), and reaches roughly $-\$1800\text{M}$ at 90 satellites where the RL policy loses only about \$110M with a gap exceeding \$1700M. In other words, operating a low-thrust servicer under myopic dispatching at high demand is far worse than deploying no servicer at all.

This collapse follows directly from the structural limitations identified in Section 3.5.1, which slow transfers turn from benign to crippling. A low-thrust transfer spans weeks to months, so every decision is a long, irrevocable commitment of the vehicle. The heuristic, whose objective accounts for neither the duration nor the propellant cost of the transfer, routinely commits the servicer to distant, marginally higher-scoring targets while closer needs accumulate. Among the three LT specific-impulse configurations evaluated, the fastest setting, ($I_{sp} = 1500$ s), yielded the highest economic performance and was therefore retained for the greedy policy. The remaining difference is thus primarily attributable to the scheduling strategy itself: the greedy policy optimizes each decision locally, whereas the RL agent can account for the longer-term consequences of target selection and service sequencing. This distinction becomes particularly important for the LT architecture, whose long transfer durations create substantial opportunity costs and make poor sequencing decisions difficult to recover from. Unable to sequence targets or anticipate upcoming demand, the heuristic lets the service backlog spiral as the fleet grows, and the forgone revenue of the queued satellites overwhelms every other term of the balance. The learned policy contains this backlog far more effectively, only dipping below the no-service baseline at 90 satellites.

For the HT servicer, the picture is entirely different: the two curves are nearly indistinguishable across the whole demand range. The heuristic even edges out the learned policy at intermediate fleet sizes (about \$1450M against \$1360M at 70 satellites), although these differences remain well

within the episode-to-episode spread, and the two policies converge again at 90 satellites (about \$1040M and \$1060M). The explanation mirrors the previous one. High-thrust transfers last hours to days, so a decision commits the servicer only briefly: a suboptimal choice is cheap and quickly corrected, and the scheduling problem nearly decomposes into a sequence of independent “serve the most valuable need now” decisions, precisely the rule the heuristic implements, built on the same forgone-revenue and expected-loss quantities as the reward function. With a single operating point, the HT configuration also offers no transfer-setting choice for a learned policy to exploit. In this regime, the RL agent matches, but does not surpass, a well-designed dispatching rule; the small residual shortfall at intermediate demand plausibly reflects the finite training budget rather than a structural limit.

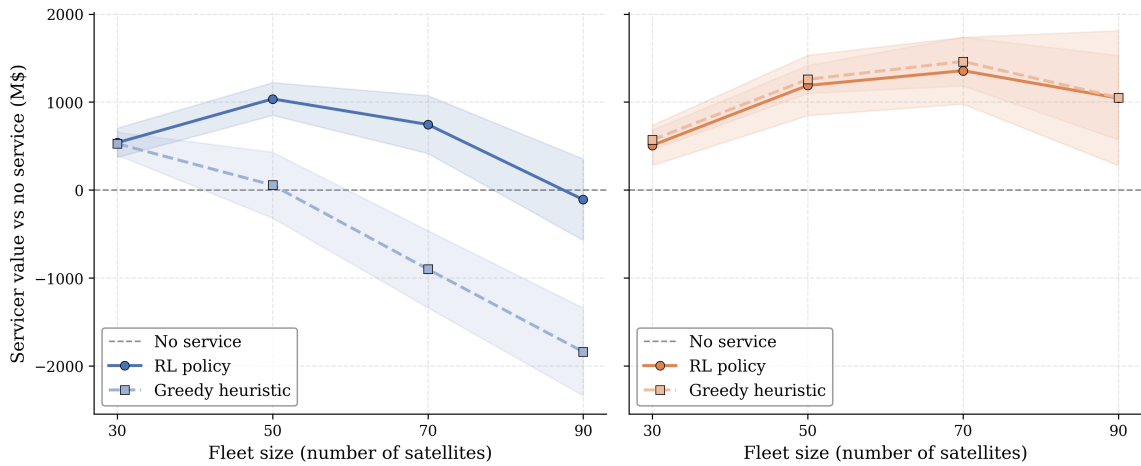


Figure 4.15: Adjusted NPV (ΔNPV , servicer value relative to the no-service baseline) as a function of fleet size, for the trained RL policy and the greedy value-at-stake heuristic of Section 3.5.1, shown separately for the low-thrust servicer (left) and the high-thrust servicer (right). Both policies drive the same servicer through the same environment and are evaluated on the same stochastic seeds (single-depot configuration, five-year horizon). Markers show the mean over the 500 evaluation episodes and the shaded bands the corresponding spread.

This near-parity, however, holds only as long as the refuelling logistics remain benign. Figure 4.16 revisits the comparison for the HT servicer in the depot case studies introduced at the end of Section 4.2.5, i.e., the geometrically separated orbits of Table 4.9 and the reduced 800 kg propellant

tank, reporting this time the full NPV of Eq. (24) for one, two, and three depots. To give the heuristic its best possible showing, its low-fuel refuelling threshold was tuned separately for each depot configuration: the threshold was swept from 10% to 95% in 5% increments, then refined in 1% steps around the best candidate, retaining the value that maximises the heuristic’s mean NPV under the same policy and economic conventions. The retained thresholds, 58%, 82%, and 74% for the one, two, and threedepot configurations, respectively, are annotated next to the corresponding points in Figure 4.16.

Even at its tuned optimum, the heuristic struggles to keep up. The gap is largest in the single-depot configuration, where the learned policy outperforms the tuned heuristic by \$783M (\$1401M against \$618M), the heuristic falling far below even the no-service baseline. The gap then narrows as depots are added, to \$99M with two depots (\$1543M against \$1444M) and a modest \$34M with three (\$1464M against \$1430M). The single-depot collapse has a clear mechanism: in many episodes, the heuristic strands the servicer early or midway through the mission, accepting an expensive outbound service transfer after which the remaining propellant no longer suffices to reach the lone depot. This is precisely the structural weakness anticipated in Section 3.5.1, i.e. a static fuel threshold is not a guaranteed-return reserve, and the exhaustive sweep shows that no static value fixes it.

The tuned thresholds themselves are diagnostic. With a single depot, every refuelling trip may require up to a 6° plane change, so refuelling often is prohibitively expensive: the optimum settles at the lowest value of the three (58%), rationing depot visits at the price of a residual stranding risk. Once depot access becomes cheap, the optimum jumps to 82% (two depots) and 74% (three): early, frequent refuelling now costs little and essentially eliminates stranding, letting the heuristic recover most, but not all, of the gap. That the best static reserve varies this widely across depot configurations, and that none rescues the single-depot case, is further evidence that a fixed refuelling rule cannot substitute for state-dependent propellant management: the learned policy folds the cost of the return trip into every targeting decision, adjusting its effective reserve to the situation at hand.

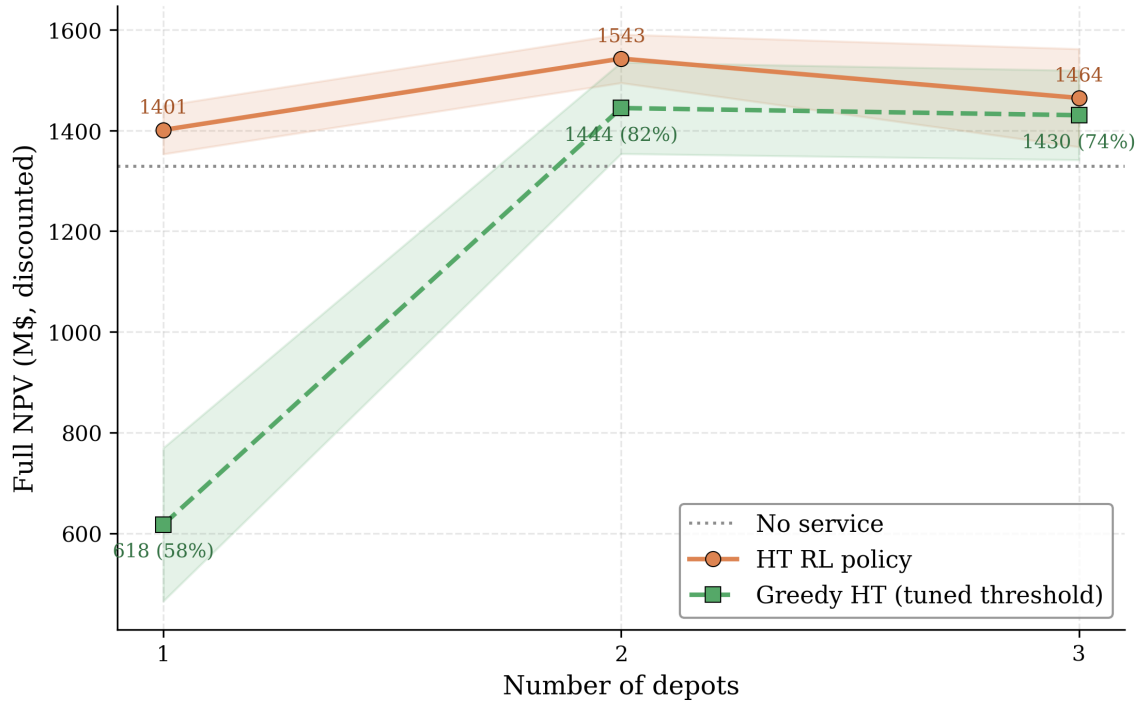


Figure 4.16: Full discounted NPV (Eq. 24) achieved by the trained HT RL policy and by the greedy value-at-stake heuristic with a per-configuration tuned refuelling threshold, as a function of the number of refuelling depots, for the depot case studies of Section 4.2.5 (modified constellation geometry of Table 4.9, two-year horizon, 24 satellites, 800 kg propellant capacity). The dotted line marks the no-service NPV, and the percentage next to each heuristic point reports the refuelling threshold retained for that configuration. Markers show the mean over the 1000 evaluation episodes and the shaded bands the corresponding spread.

Together, these comparisons answer the question posed in Section 3.5, and sharpen it. Horizon-level, anticipatory optimisation creates decisive value wherever a decision is a long or hard-to-reverse commitment. For the fuel-efficient but slow LT architecture, every transfer is such a commitment, and reinforcement learning is the difference between a profitable infrastructure and a value-destroying one at scale. For the fast HT architecture, individual transfers are cheap to correct and a transparent, training-free heuristic captures essentially all of the achievable value, but only as long as refuelling remains a minor concern. As soon as the fuel logistics become safety-critical, with

a tight tank and expensive depot access, the commitment shifts from the transfer itself to the propellant budget, and the learned policy's state-dependent reserve management again proves decisive where even an optimally tuned static refuelling rule fails.

Chapter 5

Conclusion

This thesis introduced the Integrated On-Orbit Services Scheduling optimisation (IOOSSO) framework, a deep reinforcement learning approach to the dynamic scheduling of on-orbit servicing missions in a multi-orbit environment. The servicing problem was cast as an event-driven semi-Markov decision process in which a single servicer decides, at each decision epoch, which client satellite or refuelling depot to visit next, and, in the low-thrust case, which propulsion operating point to use, while operating under fuel, time, and orbital-transfer constraints and responding to service needs that emerge stochastically over the mission. The environment couples a continuous orbital representation, on-the-fly rendezvous-arc construction for both high-thrust and low-thrust propulsion, a Weibull-based degradation and failure model, and orbital refuelling depots. A Maskable Proximal Policy optimisation agent, equipped with invalid-action masking, was trained to maximise an integrated-system net present value (NPV) reward that balances preserved fleet revenue against propellant expenditure and irreversible asset losses, and was benchmarked against a training-free greedy value-at-stake heuristic grounded in the same economic quantities. In contrast to the existing reinforcement learning literature, which has largely addressed active debris removal under simplified propulsion models, the framework treats the on-orbit servicing scheduling problem with an explicit, economically grounded comparison of high-thrust and low-thrust propulsion, stochastic degradation, and orbital refuelling infrastructure within a single reward.

The framework was evaluated on a set of MEO case studies inspired by the SES O3b mPOWER constellation, spanning one and five year mission horizons, client fleets of 30 to 90 satellites, one

to three refuelling depots, and two constellation geometries of increasing inclination separation. Several consistent findings emerged. First, the economic viability of on-orbit servicing is strongly horizon-dependent: over a one-year horizon, neither propulsion architecture recovers the cost of the servicing infrastructure, and both fall 6–7% short of the prevailing replace-on-failure strategy; over a five-year horizon, by contrast, both servicers break even within the first year of operation and create close to \$500M of value relative to that baseline, a gain of 11–12%. Second, the relative merit of the two propulsion modes depends on the level of service demand, with a crossover that occurs at a surprisingly modest fleet size. Under small demand the fuel-efficient low-thrust servicer holds a slight economic edge, since its low propellant consumption and infrequent refuelling outweigh its long transfer times; but already at a fleet size of about 34 satellites the time-efficient high-thrust servicer overtakes it. Beyond this point the low-thrust architecture saturates: its value creation peaks at 50 satellites, erodes at 70, and falls below the no-service baseline at 90 satellites, where unattended needs accumulate faster than the slow servicer can clear them, so that maintaining the servicing infrastructure then destroys value relative to simply replacing satellites upon failure. The high-thrust servicer, in contrast, remains strongly profitable across the entire demand range, peaking at 70 satellites and declining only beyond that point while staying well above the baseline, a decline the depot experiments trace to the throughput limit of a single vehicle rather than to its refuelling logistics. As demand grows, transfer speed, not fuel efficiency, becomes the decisive attribute of a servicing architecture.

A third set of results concerns the refuelling infrastructure, and qualifies the intuitive expectation that more depots make a better servicing system. In the baseline, nearly coplanar constellation, i.e., where the largest inclination separation is 3° , additional depots turn out to be economically counter-productive for both architectures. The low-thrust servicer visits a depot so rarely that the capacity unlocked by extra depots cannot offset their acquisition, launch, and station-keeping costs. Even the high-thrust agent, operating from a single depot, learns to keep its depot-access cost low by batching several services between refuelling trips, so that the propellant saved by additional depots never repays their infrastructure cost. The picture reverses once the orbital planes are geometrically separated. In a modified constellation with a worst-case inclination separation of 6° and a reduced propellant tank, the depot network becomes a first-order economic lever for the high-thrust servicer:

its value nearly triples when a second depot caps the worst-case plane change, then recedes when a third depot adds cost without reducing that worst case. Most strikingly, depot placement proves as decisive as depot count: a single, well-placed depot on the intermediate-inclination orbit captures essentially the entire value of the best two-depot configuration while carrying half the infrastructure. Taken together, these results show that propulsion choice and refuelling infrastructure cannot be designed in isolation: depots create value only where inter-orbit manoeuvres are expensive, and the most cost-effective servicing architecture is the one whose propulsion mode and depot network, in number and in placement, are jointly matched to the expected service demand and to the geometry of the constellation being served.

Finally, the comparison between the learned policies and the greedy value-at-stake heuristic delineates where anticipatory optimisation genuinely pays. For the low-thrust servicer, whose transfers span weeks to months, every decision is a long, irrevocable commitment: the myopic heuristic lets the service backlog spiral as demand grows, and its value collapses to more than \$1700M below that of the learned policy at the largest fleet size, even far worse than deploying no servicer at all. For the high-thrust servicer under benign logistics, the two policies are nearly indistinguishable: with transfers lasting only hours to days, the scheduling problem almost decomposes into a sequence of independent “serve the most valuable need now” decisions, which is precisely the rule the heuristic implements. Yet as soon as the refuelling logistics tighten, i.e. separated orbital planes, a smaller tank, a single distant depot, the heuristic strands the servicer in a substantial fraction of episodes and forfeits \$783M relative to the learned policy even after an exhaustive tuning of its refuelling threshold, showing that no static fuel reserve can substitute for the state-dependent propellant management the agent learns. Reinforcement learning is thus most valuable precisely where the scheduling problem is hardest: when decisions are long-lived commitments, as for the low-thrust servicer under growing demand, and when propellant logistics are constrained. Conversely, when transfers are fast and refuelling is cheap, a well-designed myopic rule recovers nearly all of the achievable value.

Beyond these specific findings, the study demonstrates that deep reinforcement learning, and Maskable PPO in particular, can learn adaptive and economically sensible servicing policies under realistic uncertainty, providing a unified methodology for mission planning and for the comparative

assessment of alternative propulsion, infrastructure, and scheduling-policy choices.

Several limitations temper these conclusions and point to directions for future work. The economic figures rest on publicly available, aggregate data: per-satellite revenues, replacement values, service durations, and degradation parameters are representative orders of magnitude rather than precise operational quantities, so the results are best read as relative comparisons rather than absolute valuations. The case studies are also restricted to a single servicer operating in MEO, with a propulsion mode fixed for the whole mission and idealised assumptions such as perfectly circular orbits and instantaneous satellite replacement and propellant transfer. Natural extensions include fleets of multiple, possibly heterogeneous, servicers, a direction that emerged in light of the vehicle flow saturation observed during periods of high demand; hybrid propulsion architectures that switch between high and low-thrust transfers; the joint optimisation of the number and placement of the refuelling depots, which the present results identify as a decisive economic lever; and application of the framework to LEO and GEO, as well as to cross-regime servicing missions. Calibrating the degradation and economic models against operational data, as such data become available, would further sharpen the framework's predictive value. More broadly, we believe the IOOSSO framework offers a flexible foundation for the economic assessment of on-orbit servicing as the orbital environment continues to grow in both scale and value.

Appendix A

Maskable PPO Hyperparameters

Table A.1 reports the hyperparameters of the Maskable PPO agent, common to all case studies. They were obtained through a manual (heuristic) search rather than an automated optimisation procedure. The total number of training steps is case-dependent and is reported, together with the corresponding wall-clock training times, in the main text (Section 4.2.2 and Section 4.2.3).

Hyperparameter	Value
Policy	<code>MultiInputPolicy</code> (separate actor/critic)
Hidden layers (actor π / critic V)	<code>[512, 512]</code> / <code>[512, 512]</code>
Activation function	<code>Tanh</code>
Orthogonal initialisation	Yes
Learning rate	linear schedule $1 \times 10^{-4} \rightarrow 2 \times 10^{-5}$
Rollout horizon (steps per environment)	4096
Mini-batch size	2048
Number of epochs per update	10
Discount factor γ	0.999
GAE parameter λ	0.95
Clipping range ϵ	linear schedule $0.2 \rightarrow 0.1$
Entropy coefficient c_{ent}	0.0002 (HT) / 0.01 (LT)
Value-function coefficient c_V	0.7
Maximum gradient norm	0.5
Target KL divergence	0.04
Number of parallel environments	12
Observation / reward normalisation	<code>VecNormalize</code> (clip = 10)

Table A.1: Hyperparameters of the Maskable PPO agent, tuned through a manual heuristic search. All values are shared between the high-thrust (HT) and low-thrust (LT) configurations, except the entropy coefficient, set slightly higher for the low-thrust agent to encourage exploration over its larger action space.

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